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Two-temperature effects in Hall-MHD simulations of the HIT-SI Experiment

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A two-temperature Hall-magnetohydrodynamic (MHD) model, which evolves the electron and ion temperatures separately, is implemented in the PSI-Tet code and used to model plasma dynamics in the HIT-SI experiment. The two-temperature model is utilized for HIT-SI simulations in both the PSI-Tet and NIMROD codes at a number of different injector frequencies in the $14.5 - 68.5$ kHz range. At all frequencies, the NIMROD two-temperature model results in increased toroidal current, lower chord-averaged density, higher average temperatures, outward radial shift of the current centroid, and axial symmetrization of the current centroid, relative to the single-temperature NIMROD simulations. The two-temperature PSI-Tet model illustrates similar trends, but at high frequency operation exhibits lower electron temperature, smaller toroidal current, and decreased axial symmetrization with respect to the single-temperature PSI-Tet model. With all models, average temperatures and toroidal currents increase with the injector frequency. Power balance and heat fluxes to the wall are calculated for the two-temperature PSI-Tet model and illustrate considerable viscous and compressive heating, particularly at high injector frequency. Parameter scans are also presented for the artificial diffusivity, and wall temperature and density. Artificial diffusivity and the density boundary condition both significantly modify the plasma density profiles, leading to larger average temperatures, toroidal current, and relative density fluctuations at low densities. A low density simulation achieves sufficiently high current gain ($G > 5$) to generate significant volumes of closed flux lasting 1-2 injector periods.

Keywords: plasma, MHD, self-organization, spheromak

I. INTRODUCTION

Space and laboratory plasmas are almost universally multi-scale and strongly nonlinear. Simulations can improve our understanding, but they necessarily require complex numerical models. Efficiency and model validation, balancing complexity and computational speed, is essential for simulating plasmas from the solar wind to the tokamak. Magnetohydrodynamic (MHD) codes have now been used in the plasma physics community for over 50 years, but the complexity and inherent scale separations result in many numerical codes which are specialized for the efficient computation of a specific type of experiment or plasma system. In the interest of fusion and propulsion devices, a number of codes are optimized for solving in axial and toroidal geometry[1–3].

As performance and efficiency have improved, additional physical effects have been added to the basic single-fluid, ideal MHD model, such as: resistive effects (finite dissipation), two-fluid (Hall) effects, finite Larmor radius (FLR) effects, evolution of a neutral fluid, and improved closures. Previous work on HIT-SI simulations started with a resistive single fluid model[4] followed by a model including Hall terms to capture some of the effects expected with the full two-fluid system of separate ions and electrons[5]; these models are typically referred to as resistive-MHD and Hall-MHD respectively. The importance of different physical terms in the MHD equations is interpreted from theoretical considerations[6] and the results of experimental validation[4,5,7]. Both experimentally and theoretically, two-fluid effects through the Hall terms have been shown to be important for spatial scales between the ion and electron inertial scales[8]. In HIT-SI, the ion inertial scale $d_i \approx 8$ cm is comparable with experimental length scales, such as the diameter of an injector mouth $d_{\text{inj}} \approx 14$ cm, and the characteristic magnetic scales of the spheromak $\lambda_{\text{sph}}^{-1} \approx 10$ cm, and of the injector $\lambda_{\text{inj}}^{-1} \approx 5$ cm[9,10]. Numerical models with the Hall terms significantly improve agreement with HIT-SI measurements over the resistive MHD models[5], and validate well with many bulk measurements of the experiment[7,11]. Modeling the injector geometry, especially capturing the short-lived reconnection events in the injectors, led to improved agreement with experimental measurements[12]. Additionally, from measurements of HIT-SI[13] (and a newer device called HIT-SI3[14]) and theory[9,15], ion temperatures are expected to be significantly higher than electron temperatures. Therefore, distinguishing between the ion and electron temperatures is a well-motivated change to the MHD equations, leading to new dynamics in HIT-SI simulations.

In general, investigating a collection of models and geometries of varying complexities facilitates new understanding of the plasma dynamics, as well as validation of the numerical codes. Although validation with experimental data is important for quantitative and predictive understanding of experimental results using simulations, it is not the primary goal of the present work. Rather, the present work is part of a systematic and sustained effort to investigate the complexity of the physical models, geometry, and boundary conditions required to capture the

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important dynamical processes in this device. In particular, we aim to elucidate the underlying physics through identification of important terms in the dynamics.

A. The HIT-SI experiment

HIT-SI was a laboratory device that formed and sustained spheromak plasmas for the study of plasma self-organization and steady inductive helicity injection (SIHI)[15]. It consisted of an axisymmetric flux conserver and two inductive injectors (called the X-injector and the Y-injector) mounted on each end as shown in Fig. 1. Voltage and flux circuits are wrapped around the outside of the injectors but are not pictured here. The voltage and axial flux of each injector are oscillated in phase at a frequency f_{inj} with values between $14.5 - 68.5$ kHz. The two injectors are spatially and temporally 90° out of phase and are purely inductive. The total injected power and magnetic helicity is approximately constant during the discharge and is slowly varying compared with f_{inj}. An experimental discharge begins with an initial formation period, followed by the nonlinear self-organization of the spheromak plasma, which is subsequently sustained by the injectors for the remainder of the discharge. Although the relaxation to a lower energy state forms the theoretical basis for this plasma device, the exact process (nonlinear self-organization) is not well-understood experimentally or in simulations. This nonlinear relaxation is, to an extent, examined in PSI-Tet simulations in Sec. III D. In both experiment and simulation, sustainment of the plasma is from the balance of helicity and energy injection with the corresponding dissipation in the plasma volume. Descriptions of the equilibrium profile and postulated current drive mechanism during sustainment can be found in the references on imposed dynamo current drive (IDCD)[6,16]. Detailed descriptions of the experiment can be found in Jarboe et. al.[15] and Wrobel[17]. The upgraded HIT-SI3 device is dynamically similar, but features three inductive injectors mounted solely on the top of the device.

B. Contributions of this work

A two-temperature Hall-MHD model is implemented in the PSI-Tet extended MHD code. The present work is primarily to investigate the dynamic changes with this more advanced model, rather than close experimental validation, which would require a more realistic model for the HIT-SI injectors. The PSI-Tet and NIMROD models are described in Section II and are compared with their single-temperature counterparts. In Section III, we extend the scans of the injector frequency performed in previous works[11,18] and find a number of interesting and important changes in the plasma dynamics, including: significantly hotter ions than electrons at high frequency operation, new insights into the dynamical power flows, super-linear injector impedance scaling with the injector frequency, and improved understanding of the structure of spheromak formation. When compared to the single-temperature models, the two-temperature models tend to predict larger toroidal currents, increased volume-averaged temperatures, lower chord-averaged densities, an outward radial shift of the current centroid, and vertical symmetrization of the current centroid. Interestingly, at higher injector frequencies, the two-temperature PSI-Tet model exhibits lower average electron temperatures, smaller toroidal currents, and reduced vertical symmetrization compared to the single-temperature model. Moreover, we discover that modeling the injector geometry in PSI-Tet may be important for reproducing the experimentally observed asymmetry between injector waveforms and corresponding vertical shift of the current centroid. NIMROD does not model the injector geometry but the single-temperature model still produces a slight vertical shift; this shift is not present with the two-temperature NIMROD model.

In Section IV, parameter space scans are performed for the artificial diffusivity and the density and temperature Dirichlet boundary conditions. The wall temperature scan indicates an inward shift of the current centroid and reduced volume-averaged β, consistent with current distribution changes. Scans in the density and artificial diffusivity indicate operating in a low density regime leads to a hotter plasma with higher toroidal currents and larger relative density fluctuations, consistent with experimental results and theoretical work. Low frequency, low density, high power simulations illustrate evidence of a transition to a high performance regime, with closed flux events lasting $50 - 100$ μs (or 1-2 injector periods, τ_{inj}) at spheromak gains of $G > 5$. In Section V, we conclude with a number of suggestions for future work to improve the modeling and validation capabilities of these codes. The code for the analysis of this work can be found at `https://github.com/akaptano/Two-Temperature-HITSI-Analysis` and the data that supports the findings of this study is available from the corresponding author upon request. The PSI-Tet and NIMROD codes are also available by request from Dr. Chris Hansen and `https://nimrodteam.org`, respectively.

II. DETAILS OF THE MHD MODEL

Numerical MHD modeling is a critically enabling tool for physical understanding of the HIT-SI device and design of future current drive experiments based on SIHI. We use two codes, NIMROD[1] and PSI-Tet[19], which differ primarily in their spatial discretizations. Before detailing the particulars of each code, a description the MHD model used for this work is provided.

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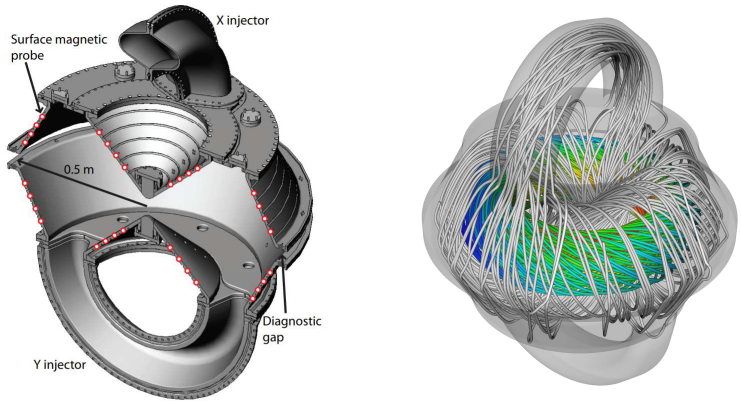


FIG. 1: Left: A cross section of the device shows the toroidal structure, the two magnetic helicity injectors, the surface probe locations, and the coordinate system. Reproduced from Wrobel et al., Relaxation-time measurement via a time-dependent helicity balance model, Physics of Plasmas, 20(1):012503, 2013, with the permission of AIP Publishing. Right: Representative equilibrium during sustainment with an injector shows an axisymmetric spheromak (rainbow) surrounded by field lines tied to the injector (gray).

A. MHD model

A first-principles model must balance efficiency and complexity, and this balance will vary depending on the physics required to resolve the dynamics of interest. The two-temperature Hall-MHD model chosen for these simulations was informed by previous work[5,18] and theoretical considerations[6,16]. This model is defined by the following evolution equations for the plasma density n, fluid velocity $\boldsymbol{u}$, ion temperature T_i, electron temperature T_e, and magnetic field $\boldsymbol{B}$,

$$\frac{\partial n}{\partial t} = -\nabla\cdot(n\boldsymbol{u})+D\nabla^2 n, \tag{1}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} = -\boldsymbol{u}\cdot\nabla\boldsymbol{u}+\frac{1}{m_i n}\left[\boldsymbol{J}\times\boldsymbol{B}-\nabla\left(n(T_i+T_e)\right)-\nabla\cdot\boldsymbol{\Pi}\right], \tag{2}$$

$$\frac{\partial T_i}{\partial t} = -\boldsymbol{u}\cdot\nabla T_i+(\gamma-1)\left[-T_i\nabla\cdot\boldsymbol{u}-\frac{1}{n}\left(\nabla\cdot\boldsymbol{q}_i-Q_i\right)\right], \tag{3}$$

$$\frac{\partial T_e}{\partial t} = -\boldsymbol{u}_e\cdot\nabla T_e+(\gamma-1)\left[-T_e\nabla\cdot\boldsymbol{u}_e-\frac{1}{n}\left(\nabla\cdot\boldsymbol{q}_e-Q_e\right)\right], \tag{4}$$

$$\frac{\partial \boldsymbol{B}}{\partial t} = -\nabla\times\boldsymbol{E}, \tag{5}$$

with terms which are different in the single and two-temperature models colored in blue. The model is completed with the following closures:

$$\mu_0\boldsymbol{J} = \nabla\times\boldsymbol{B}, \tag{6}$$

$$\boldsymbol{E} = -\boldsymbol{u}\times\boldsymbol{B}+\eta\boldsymbol{J}+\frac{1}{ne}\left(\boldsymbol{J}\times\boldsymbol{B}-\nabla nT_e\right)+f_{m_e}\frac{m_e}{ne^2}\frac{\partial \boldsymbol{J}}{\partial t}, \tag{7}$$

$$\boldsymbol{q}_s = -n\left[\chi_{\parallel,s}\hat{\boldsymbol{b}}\hat{\boldsymbol{b}}+\chi_{\perp,s}\left(\boldsymbol{I}-\hat{\boldsymbol{b}}\hat{\boldsymbol{b}}\right)\right]\cdot\nabla T_s, \tag{8}$$

$$Q_i = -\left(\nabla\boldsymbol{u}\right)^T : \boldsymbol{\Pi}-Q_{\text{coll}}, \tag{9}$$

$$Q_e = \eta\boldsymbol{J}^2+Q_{\text{coll}}, \tag{10}$$

$$Q_{\text{coll}} = 2\times10^{-14}\frac{n}{T_e^{\frac{3}{2}}}(T_i-T_e), \tag{11}$$

$$\boldsymbol{\Pi} = -\nu\boldsymbol{W} = -\nu(\nabla\boldsymbol{u}+(\nabla\boldsymbol{u})^T-\frac{2}{3}\boldsymbol{I}\nabla\cdot\boldsymbol{u}), \tag{12}$$

$$\boldsymbol{u}_e = \boldsymbol{u}-\frac{\boldsymbol{J}}{ne}. \tag{13}$$

Here m_i is the ion mass, m_e is the electron mass, e is the charge of an electron, $\gamma=5/3$ is the adiabatic index, $\hat{\boldsymbol{b}}$ is a unit vector in the direction of the magnetic field, and s indexes the fluid species, $s=i$ or $s=e$. Strictly, Eq. 1 should not contain a diffusive term. However, an artificial particle diffusivity $D=250$ m^2/s is used to prevent numerical issues with small scale density oscillations characteristic of the dynamics in HIT-SI simulations; this technique is common in simulations of dynamic plasma systems[20,21]. Unfortunately, numerical issues prevent converging to a smaller value at which the dynamics become

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insensitive to this term. However, we investigate dynamics in Section IV C with $D=50$ m^2/s, the lowest value reported so far for HIT-SI simulations. Anisotropic Braginskii thermal conduction with temperature and magnetic field dependencies[22] $\chi_{\perp,s}$ and $\chi_{\parallel,s}$, Spitzer-like resistivity[23] $\eta=\eta_0/T_e^{\frac{3}{2}}$, and constant and isotropic viscosity $\nu=550$ m^2/s are assumed. An approximate demagnetization effect, which enforces $\chi_{\perp,s}\leq\chi_{\parallel,s}$, is included in the thermal conduction coefficients because there are locations, like reconnection regions inside the injectors, where the magnetic field vanishes. The viscosity value used is approximately that obtained from Braginskii[22], assuming plasma parameters late in the discharge, and this value matches that which was used in prior studies. Previous HIT-SI simulations found that "while there are changes to the flow profile between isotropic and anisotropic viscosity models, they tend to have little influence on the bulk evolution of the plasma"[7]. The electrons are approximated in this model to receive the entirety of the ohmic heating, while only ions receive viscous heating. These choices are well-motivated because the ratio of electron to ion ohmic heat and the ratio of ion to electron viscous heat are both approximately $\sqrt{m_i/m_e}$. The heat exchange from ion-electron collisions, Q_{coll}, is an approximation obtained from the latest NRL plasma formulary[24].

The electron inertia term incorporates two-fluid effects by introducing the electron inertial scale to the evolution equation and retaining the whistler-wave dispersion cutoff at the electron cyclotron frequency, $\omega=\omega_{ce}$. In both numerical codes, there is an enhancement factor for the electron inertia term appearing in Ohm's law. Since the electron mass is artificially enhanced, high-frequency whistler waves are damped, reducing numerical stiffness in the magnetic field evolution. Moreover, to leading order $f_{m_e}m_e\dot{\boldsymbol{J}}/ne^2\approx f_{m_e}m_e\omega_{\text{inj}}\boldsymbol{J}/ne^2\propto\eta_f\boldsymbol{J}$, with η_f interpretable as additional resistivity for currents changing at a fixed frequency. In HIT-SI, for currents oscillating at the lowest frequency (14.5 kHz) used in the present work, η_f is smaller than η by a factor of 2-3. At higher frequencies, the terms are of similar magnitude. Fortunately, equilibrium and other slowly changing currents will see a smaller effective η_f. Typically $f_{m_e}=36.72$ is used because this value was found in previous work to be well-converged for simulations similar to those presented here[5,18]. A convergence study was also performed as part of this work, which showed no significant changes in the range $f_{m_e}=18.36-73.44$. These and other relevant parameters are summarized in Table I.

B. PSI-Tet

PSI-Tet is a 3D high-order finite element code that supports multi-physics models on unstructured tetrahedral grids. The grids can be generated directly from Computer-Aided Design (CAD) models and this discretization facilitates the accurate representation of complex 3D geometries like the HIT-SI device. The experimental device consists of a flux conserver constructed from 1/2" thick copper. The plasma-facing surface is coated with a thin insulating layer so that the injector operation remains purely inductive. The approximate experimental boundary conditions are

Parameter	Value [Units]
Injector Flux	0.5 [mWb]
Injector Current	8 [kA]
n_{wall}	0.75 [10^{19} m^{-3}]
T_{wall}	3 [eV]
ν	550 [m^2/s]
D	250 [m^2/s]
η_0	5.327×10^{-4} [Ωm eV$^{-\frac{3}{2}}$]
$m_i/(f_{m_e}m_e)$	100

TABLE I: Fixed parameters for PSI-Tet and NIMROD frequency scan simulations.

$$\boldsymbol{B}\cdot\hat{\boldsymbol{n}}|_{\text{wall}}=0, \qquad \boldsymbol{J}\cdot\hat{\boldsymbol{n}}|_{\text{wall}}=0. \tag{14}$$

$\boldsymbol{B}$ is the magnetic field, $\boldsymbol{J}$ is the current density, and $\hat{\boldsymbol{n}}$ is a unit normal vector to the wall. This boundary condition is enabled by the unique mixed element discretization used by PSI-Tet. In PSI-Tet, the injector flux and voltage circuits in HIT-SI are approximated by imposing $\int\boldsymbol{B}\cdot\boldsymbol{dl}=f(t)$ on independent sets of closed loops on each injector. This boundary condition fixes the current enclosed within these loops as a function of time and the values are chosen to reproduce experimental waveforms. The simulations initialize the magnetic field as a small Taylor state[10] of a few Amperes of toroidal current; this is not strictly necessary, nor does it determine the direction of the toroidal current after formation, but it tends to improve the initial numerics. More details can be found in the original works[12,19].

For the remaining fields, PSI-Tet uses Dirichlet boundary conditions equal to the initial condition for velocity, temperature, and density, with the values

$$\boldsymbol{u}_{\text{wall}}=0, \quad T_{e/i,\text{wall}}=3\text{ eV}, \quad n_{\text{wall}}=0.75\times10^{19}\text{ m}^{-3}, \tag{15}$$

These choices are typical for past HIT-SI simulations, and were chosen based on validation with experiment[4,5,18]. A more principled approach to these choices would require the evolution of a neutral fluid to model the atomic processes such as ionization and recombination, which are expected to be very important at the boundaries[25,26]. This is a subject of future work[27].

An implicit Crank-Nicolson time advance is used with a maximum time step of 40 ns, determined to be suitable by convergence studies[19]. Typical time steps are 40 ns for low frequency simulations and 10 ns for high frequency simulations. The number of grid cells used in the PSI-Tet simulations is 103,887, corresponding to a uniform grid spacing of 2.8 cm with third order basis functions. This produces an approximate resolution of 9 mm, which resolves the electron inertial scale $d_e\approx12$ mm, although most of the electron dynamics occurring at this scale are not included in Equations 1−13.

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C. NIMROD

If we are not interested in the precise effects of the injector geometry, we can gain considerable numerical efficiency by solving the evolution equations with algorithms optimized for axisymmetric geometries. NIMROD is a versatile extended MHD code used for simulating spheromaks[5,7,18,28,29] as well as many other axisymmetric or quasi-axisymmetric plasma systems[30,31]. Instead of implementing the boundary condition $\boldsymbol{J}\cdot\hat{\boldsymbol{n}}$, a thin layer of high resistivity with $\eta_{\rm wall}/\eta_{\rm plasma}\approx 10^5$ is used to impede current flow into the wall. To avoid numerical issues from a sharp jump in η, a matching boundary layer is added to the mesh so that the variation occurs within a single cell. NIMROD is also restricted to toroidally symmetric geometries, so the HIT-SI injectors are modeled as boundary conditions on the flux conserver.

The magnetic field boundary condition approximates the action of the helicity injectors through a combination of $B_\perp$ and $E_\parallel$ conditions. The spatial profile of $B_\perp$ is generated through a Grad-Shafranov solution of the injector geometry and is detailed in reference[5]. The injector voltage, which generates $E_\parallel=-\nabla V$, has the same spatial profile of $B_\perp$ and differs by a phase delay in time. As in the experiment and PSI-Tet simulations, the two helicity injectors are set to a relative phasing of 90°.

The entire wall, including the injector openings, uses a constant and uniform Dirichlet boundary condition for temperature and density with the values in Table I. A minor complication is that the two-temperature simulation at $f_{\rm inj}=14.5$ kHz was performed with $T_{\rm i,wall}=T_{e,wall}=1$ eV because of numerical issues; we will later justify in Section IV A that the dynamics are fairly insensitive to this value. More importantly, the Dirichlet temperature and density boundary conditions for the injector may lead to significantly different dynamics than those observed in PSI-Tet. These boundary conditions enforce a cold, uniform plasma across the injector mouths. Parallel heat conduction then produces cold channels of plasma in the main volume which are linked to the injectors. This is in direct contrast to PSI-Tet simulations which observe highly dynamic plasma channels in the injectors which are much warmer than the wall temperature. These NIMROD simulations can be made more dynamically similar to the PSI-Tet simulations by adjusting the Dirichlet temperature and density boundary conditions on the injector mouths, but this tends to be a heuristic procedure.

The velocity is zero at the boundary except for a constant inward flow at each injector mouth to counter-act density holes that cause numerical issues. Without this inward flow, before the spheromak has formed, a density hole often forms in the core of the device from the establishment of increased pressure near the magnetic axis. This pressure pushes particles outwards, creating a deep density hole in the core; during the formation event the density becomes so low that the Hall terms, proportional to n^{-1}, cause numerical instability. These inward flows are well-motivated by and comparable in magnitude to large flows seen in PSI-Tet simulations[11,12]. The spatial profile of the normal velocity matches the absolute value of the normal magnetic field, and the peak velocity $\approx$19.5 km/s is in approximate agreement with the flow velocity observed on the experiment with ion doppler spectroscopy[13]. Like PSI-Tet, the NIMROD initial conditions are the same as the boundary conditions for the temperatures, density, and velocity (velocity is initialized to zero everywhere except on the injector mouths), with an initial Taylor state of a few Amperes.

The average cell size in the poloidal plane was 1.8 cm and used fourth order basis functions. Eleven Fourier modes in the toroidal direction were used, corresponding to a toroidal node spacing of approximately 8.3 cm at the mid-radius of the domain (27.5 cm). Grid resolution studies for both PSI-Tet[12,19] and NIMROD[5,18] have shown convergence of results at these resolutions. Lastly, NIMROD solves the same system of equations that PSI-Tet does, with the exception of a divergence cleaning term added to the magnetic field evolution[1]. Before moving on to the results, we identify terms in the power flows in order to disambiguate various physical effects.

D. Power flows

Significant differences in ion and electron temperatures can often be explicated by an examination of the heating and loss terms. The injector voltage circuit is responsible for most of the experimental power input, but in PSI-Tet only the current waveform of this circuit is known. For our purposes, the consequence is that the total injected power cannot be calculated directly in PSI-Tet. Instead, the total injector power is approximated from a power balance of the total time rate of change of the total energy with the ion and electron heat fluxes to the wall through

$$P_{\rm inj}+\oint_\Omega(\boldsymbol{q}_i+\boldsymbol{q}_e)\cdot\boldsymbol{d\Omega}= \frac{d}{dt}\int_V\left[\frac{B^2}{2\mu_0}+\frac{n(m_iu^2+m_eu_e^2)}{2}+\frac{n(T_i+T_e)}{\gamma-1}\right]dV. \tag{16}$$

We also track individual thermal energy flow terms

$$\underbrace{\frac{\partial}{\partial t}\int_V\frac{n(T_i+T_e)}{\gamma-1}dV}_{\text{thermal power}}=\underbrace{\oint_\Omega(\boldsymbol{q}_e+\boldsymbol{q}_i)\cdot\boldsymbol{d\Omega}}_{\text{heat flux to wall}} + \underbrace{\int_V\nu(\nabla\boldsymbol{u})^T:\boldsymbol{W}dV}_{\text{viscous heat}}+\underbrace{\int_V\eta J^2dV}_{\text{ohmic heat}}+\underbrace{\frac{1}{\gamma-1}\int_V(T_i+T_e)D\nabla^2ndV}_{\text{diffusive heat}} \underbrace{-\frac{1}{\gamma-1}\int_V\left[(T_i+T_e)(\boldsymbol{u}\cdot\nabla n+\gamma n\nabla\cdot\boldsymbol{u})+\frac{(\gamma-1)T_e}{ne}\boldsymbol{J}\cdot\nabla n\right]dV}_{\text{density advection and compressive heat}} \underbrace{-\frac{1}{\gamma-1}\int_V\left[n\boldsymbol{u}\cdot(\nabla T_i+\nabla T_e)-\frac{1}{e}\boldsymbol{J}\cdot\nabla T_e\right]dV}_{\text{temperature advection}}. \tag{17}$$

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Changes to the above relation from upwinding[32], applied to the density and temperature evolution equations, are included and contribute negligible heating power. Heat flow is introduced by the artificial diffusivity term, the unphysical term in Eq. 1 used only to stabilize the numerical evolution. This artificial heat flow accounts for approximately 5% of the total power flowing at $f_{\rm inj}=14.5$ kHz and approximately 3% at $f_{\rm inj}=68.5$ kHz, a small but non-trivial contribution.

III. FREQUENCY SCAN

Driven nonlinear systems tend to exhibit resonances and reproduce harmonics or quasi-harmonics of the driving frequency[33]; simulating HIT-SI with different injector frequencies provides understanding about these qualitative behavior changes. Experimental trends in injector frequency indicate that higher frequency operations tend to exhibit higher plasma impedance in the injectors, increased volume-averaged β, reduced chord-averaged density fluctuations from interferometry, and, to a lesser extent, larger current gain $G=I_\phi/I_{\rm inj}$. Previous work[7,11,18] with the single-temperature models in NIMROD and PSI-Tet has indicated qualitative agreement with all of these trends when the injector frequency is increased from 14.5 kHz to 68.5 kHz; the two-temperature model facilitates further exploration and understanding of these qualitative trends. For the remainder of the paper, all volume-averaged quantities will be denoted like $\langle\beta\rangle$, as will the chord-averaged density $\langle n\rangle$ which is used in place of the volume-averaged density throughout the paper. The volume averages of the various plasma quantities and power flows use the entire plasma volume, including the injectors. The toroidal current and current centroid are obtained from averages over a number of surface magnetic probes and will be denoted $\bar{I}_\phi$ and $(\bar{R},\bar{Z})$, respectively.

Most of this analysis will be concerned with the plasma dynamics observed in these simulations, but qualitative comparisons with experimental trends will also be made. It is important to note that these simulations use typical parameters for a HIT-SI discharge at 68.5 kHz, i.e. at relatively low density and power. Furthermore, because the X and Y injectors are operated with a 90° phase shift with respect to one another, the total power injected in experimental HIT-SI satisfies

$$\begin{aligned} P_{\rm inj} &= V_{\rm x,inj}I_{\rm x,inj}+V_{\rm y,inj}I_{\rm y,inj} \\ &= V_{\rm inj}I_{\rm inj}[\cos^2(\phi)+\sin^2(\phi)] = V_{\rm inj}I_{\rm inj} = Z_{\rm inj}I_{\rm inj}^2 \propto f_{\rm inj}I_{\rm inj}^2, \end{aligned} \tag{18}$$

where $Z_{\rm inj}$ is the plasma impedance in the injector and $Z_{\rm inj}\propto f_{\rm inj}$ is justified later in Eq. 22. These simulations keep the injector current and flux waveform amplitudes fixed, so that if the same scaling exists in the simulations, the power injected increases linearly with $f_{\rm inj}$. In fact, we will see in Section III C that the simulations produce injector power scaling with $f_{\rm inj}$ that is slightly super-linear. While the waveform amplitudes are constant, the relative phases between the waveforms are changed to match the average phases used for high performance HIT-SI experimental discharges at each frequency, and this may also affect the injector impedance scaling with injector frequency; more details on the how the phases were chosen can be found in previous work[19].

A. Ion and electron temperatures

For low density or low temperature plasmas, the slow ion-electron collision rates can result in separate temperatures for the two species on the timescales of interest. For the HIT-SI experiment, which sustains plasmas for 1-2 milliseconds, the approximate thermalization time $\tau_{ei}\approx 100-200\ \mu$s $>\tau_{\rm inj}$ for all the injector frequencies investigated here. This suggests that injector heating occurs faster than collisions can equilibrate the species, leading to potentially substantial differences in electron and ion temperatures. Therefore we expect that separate ion and electron temperature evolutions are important for understanding the specific temperature profiles (affecting η and β), as well as the flow of plasma energy and heat through the system.

The two-temperature model shows increasing differences between $\langle T_e\rangle$ and $\langle T_i\rangle$ as the frequency increases in Fig. 2. At 14.5 kHz $\langle T_e\rangle\approx\langle T_i\rangle$. However, at 68.5 kHz $\langle T_i\rangle$ and $\langle T_e\rangle$ match closely with the experimental measurements of $T_e\approx 5-10$ eV and $T_i\approx 20-30$ eV[13,34]. Thomson scattering measurements on the newer HIT-SI3 device at 14.5 kHz also indicate electron temperatures of $5-10$ eV[14], in excellent agreement with the $\langle T_e\rangle$ calculated here. Volume-averaged electron temperature is fairly insensitive to injector frequency, while $\langle T_i\rangle$ shows a strong and approximately linear scaling with injector frequency. The mild increase for $\langle T_e\rangle$ can be partially explained through increases in the ohmic heating and electron heat flux to the wall. Collisional heating with the ions is in the tens of kilowatts and therefore plays essentially no role in the electron heating. Ohmic heating and the total electron heat flux to the wall both increase by a factor of three or four from low to high frequency, restraining volume-averaged electron temperature to a modest increase at high frequency.

The larger volume-averaged ion temperatures at high frequency operation can be explained from the corresponding increased viscous heating through the following arguments. This anomalous ion heating is the conversion of magnetic energy to ion thermal energy through magnetic reconnection. Magnetic reconnection primarily converts the magnetic energy to ion thermal energy through shock or viscous heating of the reconnection outflow[35]. Work using a stochastic ion heating model for the reconnection events indicates the fractional energy transfer is proportional to the species mass, providing a possible explanation for the hot ions and cold electrons[36]. Because most of this heating is lost through stochastic field

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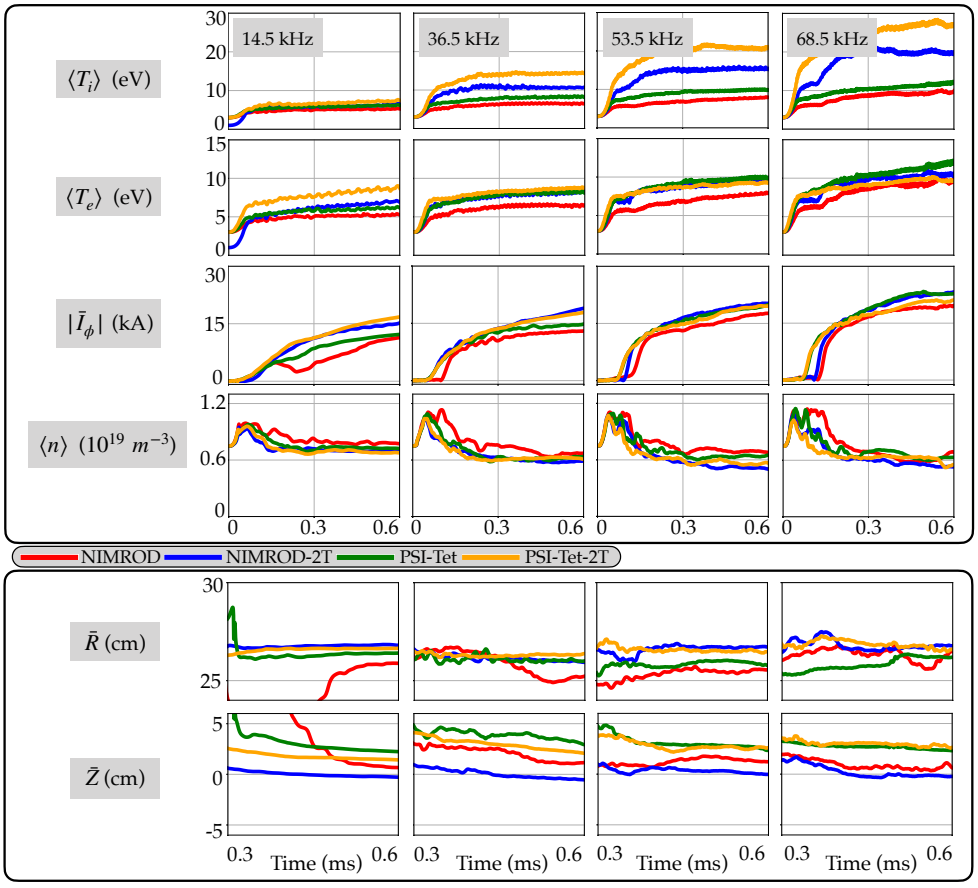


FIG. 2: Summary of bulk metrics, comparing PSI-Tet and NIMROD single and two-temperature models. Density illustrations represent chord-averaged density $\langle n \rangle$ obtained from synthetic far-infrared interferometry. Magnitude of the toroidal current $|\bar{I}_\phi|$ and current centroid $(\bar{R}, \bar{Z})$ are calculated as averages of the four poloidal surface arrays.

lines in HIT-SI simulations, the thermal energies for both species builds up until the transport to the wall equals the rate of conversion from magnetic energy to thermal energies. The increased temperatures at higher frequency operation are likely a consequence of increased magnetic reconnection (and therefore increased viscous and ohmic heating) at higher injector frequencies, since field reversals in the injectors are happening more frequently.

A likely source of additional heating is through the compressive heating $\propto n(T_e+T_i)\nabla\cdot\boldsymbol{u}$. Once the ions are warm, the compressive heat preferentially goes to the ions through the linear dependence on temperature, and Fig. 3 indicates that the total compressional heat $\propto n(T_e+T_i)\nabla\cdot\boldsymbol{u}$ is comparable to viscous and ohmic heating. Compression also tends to be very high in the injectors during a field reversal. This is both because the plasma in the injectors tends to get relatively hot (see, for instance, Fig. 4) and because velocity gradients become very large during field reversals as plasma is accelerated from the walls to carry in oppositely directed flux. This creates an outer shell of plasma with flux in one direction, which compresses an inner channel of plasma flux in the opposite direction until it becomes unstable and global reversal occurs through reconnection[19]. The early and large (larger than ohmic for the simulations with $f_{\text{inj}} \geq 36.5$ kHz) viscous heating of the ions facilitates the compressive heat to preferentially heat the ions even further. The temperature profiles are illustrated

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for the two-temperature PSI-Tet simulation at high frequency operation by the 2D contours in Fig. 4.

To understand why the PSI-Tet two-temperature model exhibits larger $\langle T_e \rangle$ than the single-temperature model at low frequency and smaller $\langle T_e \rangle$ at high frequency, note that, at low frequency, the ohmic heating is greater than the viscous heating. The single-temperature model shares the ohmic heating with the ions, reducing the amount of heat to the electrons. At high frequency, the viscous heating is shared between the electrons and ions, so that the single-temperature model now overheats the electrons. Although the NIMROD two-temperature model still produces slightly larger $\langle T_e \rangle$ than the single temperature model at high frequency, the trend with frequency is the same as the two-temperature PSI-Tet simulations (the gap between $\langle T_e \rangle$ in the single and two-temperature NIMROD simulations has shrunk as the frequency increases). It is unclear why the transition has not yet happened for the high-frequency NIMROD simulations.

B. Current centroid

The current centroid is an important quantity for modeling magnetic equilibria and understanding the magnetic topology in HIT-SI. The toroidal current $\bar{I}_\phi$ and current centroid $(\bar{R}, \bar{Z})$ are calculated as averages of the results from each of the four poloidal surface probe arrays in order to be consistent with the experimental analysis. The experimentally measured current centroid is taken as the weighted average of the poloidal field measurements from each of the four poloidal surface magnetic probe arrays[17]

$$\bar{R} = \frac{1}{4} \sum_{k=1}^{4} \frac{\sum_{i=1}^{16} R_{ik} B_{\theta,ik}}{\sum_{i=1}^{16} B_{\theta,ik}}, \tag{19}$$

$$\bar{Z} = \frac{1}{4} \sum_{k=1}^{4} \frac{\sum_{i=1}^{16} Z_{ik} B_{\theta,ik}}{\sum_{i=1}^{16} B_{\theta,ik}}, \tag{20}$$

where the index k is summing over the four poloidal arrays and the index i is summing over the 16 magnetic probes in each array. The uncertainties in the radial and vertical locations are also averaged over the four probe arrays and are quantified by the variances $\bar{\sigma}_R$ and $\bar{\sigma}_Z$. Experimental measurements have indicated an outward radial shift and vertical symmetrization of the current centroid at high frequency in HIT-SI[37]. Previous work comparing single-temperature and constant-temperature simulations did not find evidence of robust changes for different injector frequencies[11]. The current centroids computed using the two-temperature models are illustrated in Fig. 2. Compared to their single-temperature counterparts, they tend to exhibit a small outward radial shift. Both two-temperature models tend to produce vertical symmetrization of the current centroid compared to single temperature models, although high-frequency PSI-Tet simulations shift very little, or even reverse this trend. As before, there is no consistent evidence of symmetrization or outward radial shift as the injector frequency increases.

An interesting finding is that the PSI-Tet models tend to see the vertical component of the current centroid shifted upwards $2-4$ cm from the midplane, although $\bar{\sigma}_Z$ has a similar magnitude. The change in dynamics from the single to two-temperature PSI-Tet model appears to reduce the upward shift of $\bar{Z}$ at low injector frequency, but has negligible change at high frequency. The vertical shift observations in PSI-Tet appear to be the consequence of an asymmetry between the amount of power input by the two injectors in PSI-Tet[19]; injected power asymmetry is also observed experimentally. The current waveforms for the flux circuit tend to be larger for the injector on the bottom of the machine to produce the same flux in both circuits. This asymmetry depends in principle on the nonlinear plasma dynamics inside of each injector, and it is unknown how the stronger injector is determined. Experimental work suggests that the asymmetry is determined primarily by the toroidal current direction through the injector-spheromak coupling[38] but the two-temperature PSI-Tet simulations presented here with opposite toroidal current direction exhibit the same injector asymmetry. In PSI-Tet simulations the bottom injector will tend to input more power, and therefore perturb the system preferentially in the upwards direction. This is directly reflected in PSI-Tet by the vertical component of the current centroid; NIMROD only produces a slight current centroid asymmetry with the single-temperature model, suggesting that capturing the injector power asymmetry correctly may require modeling the dynamics inside the injectors.

The vertical shift of the current centroid is observed to a small extent (≈ 1 cm) experimentally[37]. The smaller experimental shift is likely because an increase in one of the injector voltage waveforms tends to occur with a corresponding decrease in the same injector current waveform, leading to approximately equal power from each injector despite the asymmetric waveforms. Dealing with this subtlety would require a more realistic model of the experimental circuit. Nonetheless, this effect provides an additional metric for experimental validation and an interesting direction for future experimental work, i.e. intentional asymmetric operation of the injectors to investigate the current centroid dependence.

The experimentally observed shift in the radial component of the current centroid has been postulated as a Shafranov shift from confined plasma pressure in the interior of the device. However, the simulations presented in this frequency scan do not exhibit evidence of closed flux surfaces or confined pressure. Thus, not surprisingly, the simulations indicate that the current centroid does not consistently shift outwards with higher frequency, even with significantly larger $\langle \beta \rangle$. The volume-averaged plasma β is pictured in Fig. 3 and defined here through

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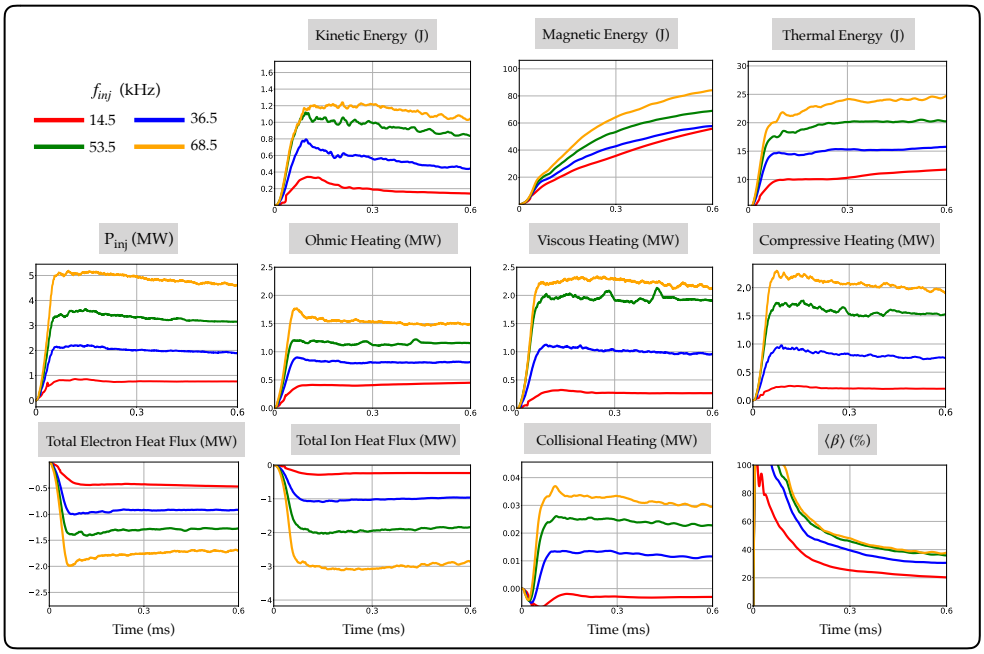


FIG. 3: Summary of volume-integrated energies and powers for two-temperature PSI-Tet simulations.

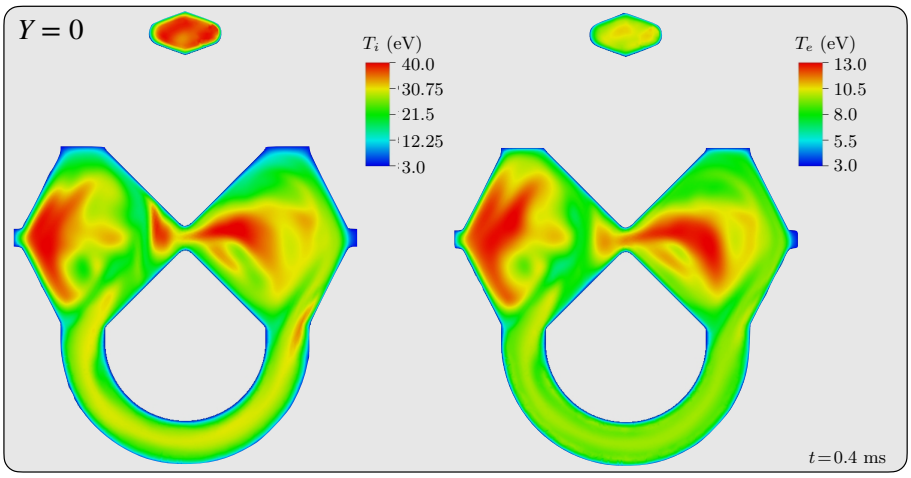


FIG. 4: 2D contours of T_i and T_e in two-temperature PSI-Tet simulations with $f_{\text{inj}} = 68.5$ kHz indicate hot injectors and similar spatial distributions for T_i and T_e.

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the plasma pressure $p=n(T_i+T_e)$ as

$$\langle\beta\rangle=\frac{\int_V(p-p_{\rm wall})dV}{\int_V\frac{B^2}{2\mu_0}dV}. \quad (21)$$

The lack of an outward shift of $\bar{R}$ as the frequency changes is consistent with single-temperature PSI-Tet observations in previous work[11]. The observed outward shift of $\bar{R}$ in the two-temperature PSI-Tet simulations is primarily a due to changes in the spatial distribution of T_e, leading to changes in the parallel current distribution through η, and thus alter the force-free component of the equilibrium. This is justified in similar simulations in Sec. IV A.

The two-temperature PSI-Tet observations for $\langle\beta\rangle$ in Fig. 3 indicate a large increase from 14.5 kHz to 36.5 kHz (perhaps corresponding to a transition from lower to higher average $\langle\beta\rangle$ as has been demonstrated elsewhere[18]) and diminishing increases from 36.5 kHz to higher frequencies.

C. Injector impedance

Experimental data and imposed dynamo current drive[6,16] predict that the injector impedance approximately satisfies

$$Z_{\rm inj}=C_1\mu_0R_0\left(\frac{1}{8\pi ea^3}\frac{\lambda_{\rm inj}}{\lambda_{\rm sph}}\frac{I_\phi}{n}+2\pi C_2f_{\rm inj}\right). \quad (22)$$

For NIMROD and PSI-Tet simulations the second term on the right-hand side is dominant[18], and so we further approximate to

$$Z_{\rm inj}\approx 2\pi CR_0\mu_0f_{\rm inj}. \quad (23)$$

$R_0=0.5$ m and $a=0.25$ m are the major and minor radius of the HIT-SI device, and C_1, C_2, $C=C_1C_2$ are fitting parameters. The $Z_{\rm inj}$ scaling with the frequency (Eq. 23) for two-temperature PSI-Tet simulations is indicated in Fig. 5 and indicates an average $C\approx 0.23$, in rough agreement with the two-temperature NIMROD simulations with $C\approx 0.125$. A lower impedance in NIMROD is reasonable, as the injector regions are not included. This means that the power used to generate magnetic fields in the injector volumes are not captured in the NIMROD simulations.

Fig. 5 indicates that this relationship is slightly superlinear in two-temperature PSI-Tet simulations but only linear in two-temperature NIMROD simulations (note that the 14 kHz NIMROD simulation is somewhat of an outlier, perhaps because this is the simulation with 1 eV walls, so we have not included it in the average C). The physical cause of the additional scaling in PSI-Tet does not appear to correlate with average $|\boldsymbol{J}|/n$ as suggested by IDCD. Due to the continuous evolution of quantities and their profiles in these simulations as the injector frequency changes, an alternative physical correlation could not be found.

A natural extension of this analysis is an examination of the scaling of the current gain $G=I_\phi/I_{\rm inj}$. Helicity balance models[34,39] indicate that for fixed injector waveforms and steady-state operation

$$G\propto\sqrt{\tau_{\rm L/R}Z_{\rm inj}}\propto\sqrt{T_e^{\frac{3}{2}}f_{\rm inj}}, \quad (24)$$

where the second scaling follows from this analysis and $\tau_{\rm L/R}$ is the resistive decay time. However, these simulations are not at steady-state operation at $t=0.6$ ms, as can be seen straightforwardly in the continued growth of the toroidal currents in Fig. 2. Further work should examine the scaling of gain and injector impedance at steady-state with the two-temperature models.

D. Spheromak formation

The nonlinear relaxation event, the interval during which the plasma self-organizes into a spheromak plasma, has been analyzed extensively. However, the exact timing of this event, and the process by which the resulting toroidal current direction is determined, are not well-understood experimentally or theoretically. Recently, work with reduced order models using dynamic mode decomposition (DMD)[40–42] identified previously undiscovered large-scale magnetic structures with $n_\phi=2$ toroidal fourier structure, oscillating at $2f_{\rm inj}$ during sustainment[43]. That work was performed in NIMROD on a larger and significantly hotter version of HIT-SI with a uniform and constant temperature and density Hall-MHD model[29].

At high frequency, NIMROD[18] and PSI-Tet single and two-temperature simulations all indicate the presence of a structure of two oppositely oriented flux tubes during spheromak formation which matches the spatio-temporal dependence identified in the DMD work. This structure also briefly appears during the formation event for low frequency HIT-SI simulations using the two-temperature PSI-Tet model. The structure is visualized using B_z at the $Z=0$ midplane for both low and high frequency two-temperature PSI-Tet simulations in Fig. 6. A 3D isosurface of $\lambda=\mathbf{J}\cdot\mathbf{B}/|\mathbf{B}|^2=13$ in Fig. 7 indicates that there are paths of constant λ which connect the two injectors, and suggest current loop formation[21] in the center of the device. These paths are reminiscent of the flux tubes, suggesting that the tubes may be generated by the magnetic field parallel to the current in these 3D pathways. Once the formation phase of the discharge gives way to the sustainment phase, these flux tubes are more difficult to identify, as the leading order magnetic field dependence now comes from the self-organized spheromak in the center of the device.

This observation is notable because the HIT-SI experiment tends to produce both negative and positive toroidal discharges and this parity choice appears to depend on a number of factors, including the phases of the injector waveforms during the time of formation[18].

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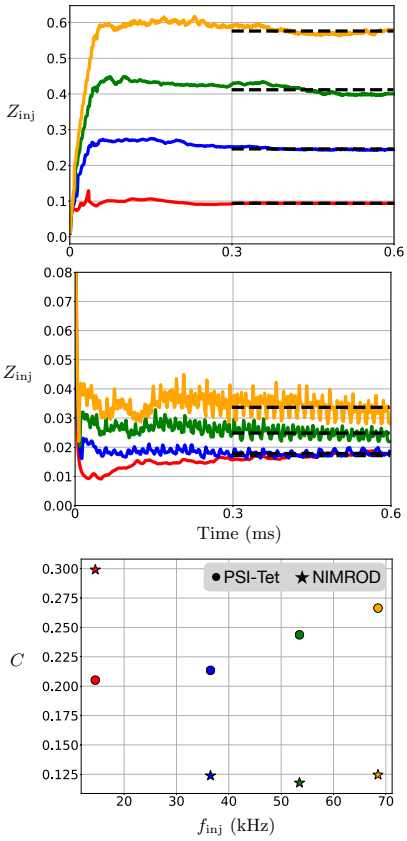
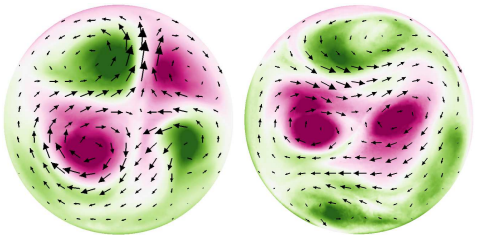


FIG. 5: Top: Eq. 23 (dashed black lines) captures the first order time evolution of $Z_{inj}(t)$ (color lines) for two-temperature PSI-Tet simulations. Middle: Equivalent to the top illustration for two-temperature NIMROD simulations. Bottom: Best fit values for C, corresponding to the black lines from above, indicate averaged $C \approx 0.23$ and $C \approx 0.125$ for two-temperature PSI-Tet and NIMROD simulations, respectively.

Previous studies have found that fixed parameter discharges operating a single injector always form and sustain spheromaks with toroidal current parity determined by the sign of the injected helicity[44,45] but more detailed explanations for this behavior have been elusive. Moreover, HIT-SI simulations in PSI-Tet with these fixed parameters always produce negative toroidal current; equivalent simulations for the HIT-SI3 device, which has a different injector geometry, produce positive toroidal currents. This parity can often be switched in the simulations by changing the relative phases of the injector



(a) $f_{inj} = 14.5$ kHz (b) $f_{inj} = 68.5$ kHz

FIG. 6: Two-temperature PSI-Tet simulations: Contours of B_z with limits ± 100 Gauss and vector plots of $\boldsymbol{B}$ illustrated at time snapshots directly before spheromak formation, in the $Z = 0$ midplane. The flux tubes form a $n_\phi = 2$ toroidal fourier structure, and oscillate at approximately $2f_{inj}$. During spheromak formation, the flux tubes merge to determine the direction of the toroidal current.

waveforms. Different parameter regimes indicate different parity preferences.

In the simulations presented here, at the time of spheromak formation, the two closest flux tubes merge in the center, while the other two spread out and merge along the edge. Which flux tubes merge in the center determines the direction of the toroidal current. All of these observations are consistent with the interpretation of each injector driving a flux tube pair, with parity fixed by the sign of helicity. During operations with both injectors, a quasi-random process then selects which flux tube pair merges and determines the sign of toroidal current. This quasi-random process likely depends on a number of nonlinearities, as different parameter spaces indicate different timing and sign of the toroidal current. Future work could investigate how these formation structures change with different injector phasing.

IV. PSI-TET PARAMETER SCANS

Investigating the parameter space of the magnetohydrodynamic models presented here is essential; fixed numerical terms require convergence studies to understand their impact on simulations and physical parameters should be scanned in order to understand the possible range of results from the experimental uncertainty in the measured values.

With the two-temperature PSI-Tet model, a number of parameter scans were performed to investigate the plasma dependence on these values. The wall temperature, wall density, and artificial diffusivity scans are performed only for $f_{inj} = 14.5$ kHz. With the exception of the wall density scan, which is performed at high injector power to investigate the possibility of closed flux

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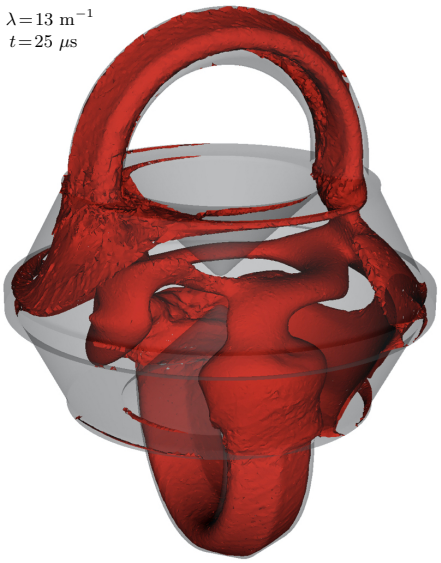


FIG. 7: A 3D isosurface of $\lambda = 13$ at $f_{\text{inj}} = 14.5$ kHz, before spheromak formation, indicates paths connecting opposite injector mouths and suggests the formation of a current loop.

formation, the simulations in this section use the parameters in Table I (although the parameter being scanned is changing from the table value). Previous NIMROD work with HIT-SI simulations has scanned the viscosity and indicated only small changes from the value of the viscosity and choice of isotropic or anisotropic viscosity[7]. However, scans performed in NIMROD may not see dependencies on the injector geometry and associated field reversals. Future work in PSI-Tet should explore this possibility.

A. Wall temperature

The exact temperature boundary condition is unknown on the HIT-SI experiment. More sophisticated first-principles modeling of the temperature boundary condition would necessitate the evolution of a neutral fluid, as plasma-wall interactions involve recombination and other atomic processes which may strongly alter the density and temperature near the boundaries. However, scanning the wall temperature provides understanding about how the internal plasma dynamics are affected and provides a sense of which value best validates with experimental observations.

Three different wall temperatures $T_{\text{i,wall}} = T_{\text{e,wall}} = 1$, 3, and 10 eV, were investigated. The only changes of any significance are shown in Fig. 8 and indicate that as the wall temperature is increased, a large inward shift of the current centroid occurs. The changes from 1 to 3 eV walls are particularly small, justifying our comparison between 1 eV and 3 eV wall temperature simulations in Sec. III. In Fig. 9, time-averaged spatial profiles of the electron temperature and J_y are compared between the 1 eV and 10 eV cases. It appears that changes to the spatial distributions in T_e primarily lead to changes in the parallel current distribution through η. This likely produces a modified force-free part of the equilibrium. In contrast, $J_\perp$ is quite similar as the wall temperature changes, as we might expect of a plasma simulation without significant indication of confined pressure. Therefore, the current distribution changes are posited to be responsible for the inward shift of the current centroid. The shift is inward because η decreases rapidly as $T_{e,wall}$ increases, facilitating stronger parallel current flows.

B. Wall density

Experimentally, HIT-SI discharges tend to reach higher current gains during low-density discharges[13,18]. To investigate the density dependence in simulations, wall densities $n = 10^{19}$ m^{-3}, 2.58×10^{19} m^{-3}, and 5.16×10^{19} m^{-3} are scanned at $f_{\text{inj}} = 14.5$ kHz. To better approximate experimental conditions, injector current and flux waveform amplitudes are increased by a factor of 2.6 as compared to the other 14.5 kHz simulations shown in previous sections, leading to higher input powers of $4-8$ MW. This is comparable with the low end of experimental discharges at low frequency, which typically input $5-15$ MW of injector power.

Fig. 10 indicates that the toroidal current is strongly dependent on the density. At low density, $\langle T_i \rangle$ exceeds 50 eV, and toroidal current is above 100 kA and growing. Despite large gains in the temperature, $\langle \beta \rangle$ decreases, indicating the magnetic field strength has increased significantly faster than the pressure. The lowest density simulation $\langle \beta \rangle$ falls to near the Mercier limit for the HIT-SI flux conserver[46,47]. The large magnetic field strength, the spheromak gain of $G > 5$, and increased fluctuations on the average temperature evolution all indicate possible closed flux activity. Previous work found transient closed flux events from viscous effects but this occurred only for magnetic Prandtl number $Pm = \mu_0 \nu / \eta \gg 100$. In contrast, this work has Pm an order of magnitude smaller and exhibits significantly larger and more sustained closed flux periods. A Poincaré plot movie in Fig. 11 for $\boldsymbol{B}$ illustrates closed flux regions lasting $50-100$ μs, or $1-2$ injector periods. This movie illuminates considerable variation in the closed flux surfaces; the flux surfaces vary from symmetric states to highly asymmetric states exhibiting complex magnetic island structures.

The large increase in viscous heating (and therefore ion temperature) at low density can be mostly accounted

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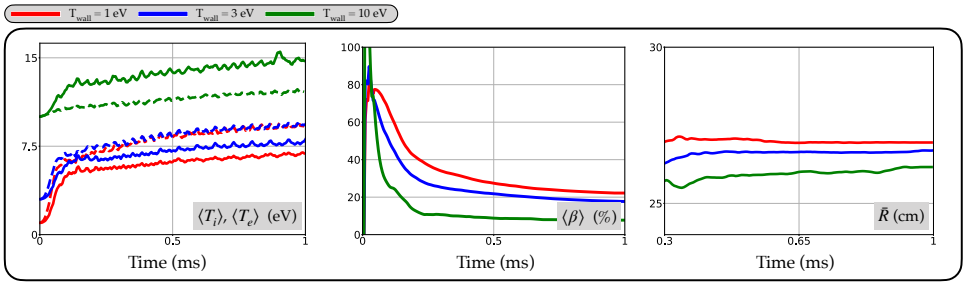


FIG. 8: Time evolution of important quantities for the wall temperature scan of two-temperature HIT-SI simulations with $f_{\rm inj}=14.5$ kHz. $\langle T_i\rangle$ and $\langle T_e\rangle$ are in solid and dashed lines, respectively.

for by the corresponding large increase in injector power. However, the simulation with $n=2.58\times10^{19}$ m^{-3} actually produces a slight decrease in the injector power compared to when $n=5.16\times10^{19}$ m^{-3}. A similar trend is seen in the compressive heating, although it exhibits a complicated temporal dependence not observed in any other of the simulations in this work. All of these observations suggest a threshold at a low enough density and high enough input power where the overall performance of the device increases sharply. An experimental study at 36.5 kHz operation did not find evidence of substantial toroidal current increases with low density deuterium[44], although the total input powers were less than 4 MW for all discharges examined. The record for experimental HIT-SI gain was $G=3.9$, but this high frequency discharge also had only a few MW of input power[37], so the $G>5$ regime has not been explored experimentally. Therefore, these observations at low density and high input power may indicate a route toward optimizing for higher performance experimental discharges. One complication in this route is that there are often large radiative losses in HIT-SI discharges, which are not modeled in the simulation. These losses may prevent the plasma from reaching the requisite viscous, ohmic, and compressive heating necessary for this higher performance regime. These findings at low density, low frequency, and high power operation merit future work on the density boundary conditions and profiles used for HIT-SI simulations.

C. Particle diffusivity

From the previous analysis, the sensitivity of the spheromak performance to the density suggests that artificial particle diffusivity may change the dynamics considerably. Unfortunately, the artificial particle diffusivity in the continuity equation cannot be completed removed because it is found empirically to be necessary for the stability of the Hall-MHD model; it is tolerable if it can be converged down to a small value where it has negligible effects on the simulation. However, previous work[18] only managed to reduce this value to approximately $D=250$ m^2/s. Here we report successful runs down to $D=50$ m^2/s in PSI-Tet, which have been reproduced in NIMROD using additional hyper-diffusivity. We also compare with a run with $D=1000$ m^2/s to understand the physical effects of this stabilizing term to regimes used in prior work[7]. We are unable to converge this value until the plasma dynamics are completely insensitive to it, but this comparison provides understanding about how diffusion affects the dynamics in our simulations. The quantities which indicated significant changes are summarized in Fig. 12.

In PSI-Tet and NIMROD this term is needed to avoid overshoot with grid-scale sharp features, generally near the wall or reconnecting regions. The two-temperature PSI-Tet simulations presented here have $\langle n\rangle$ which only mildly varies from $D=1000$ $m^2/s\rightarrow250$ m^2/s but is significantly lower at $D=50$ m^2/s. This reduction in chord-averaged density produces an overall increase in performance (higher toroidal current and average temperatures) consistent with the wall density scan. The volume-averaged temperatures increase significantly, and $\langle T_i\rangle>\langle T_e\rangle$ for $D=50$ m^2/s. Relatively unchanged thermal pressure, with significantly larger magnetic pressure at low diffusivity, leads to reduced $\langle\beta\rangle\approx10\%$.

Experimental interferometry exhibits large and rapid oscillations which are considerably larger than those observed in previous work or in this analysis; artificial diffusivity has been postulated as a possible explanation for this discrepancy. The density fluctuations $\langle\delta n\rangle$ shown on the raw density signal in Fig. 12 grow approximately proportionally as the diffusivity is decreased, and the relative density fluctuations $\langle\delta n\rangle/\langle n\rangle$ also increase significantly. Additionally, Fig. 13 indicates that in the low diffusivity case, there can be spatial variations in the density of almost three orders of magnitude. These large gradients in the density can lead to numerical instability. Since both absolute and relative density fluctuations decrease as D decreases, it appears likely that further convergence of $D\rightarrow0$ will validate better with the large experimental oscillations. Another possibility is that matching the ex-

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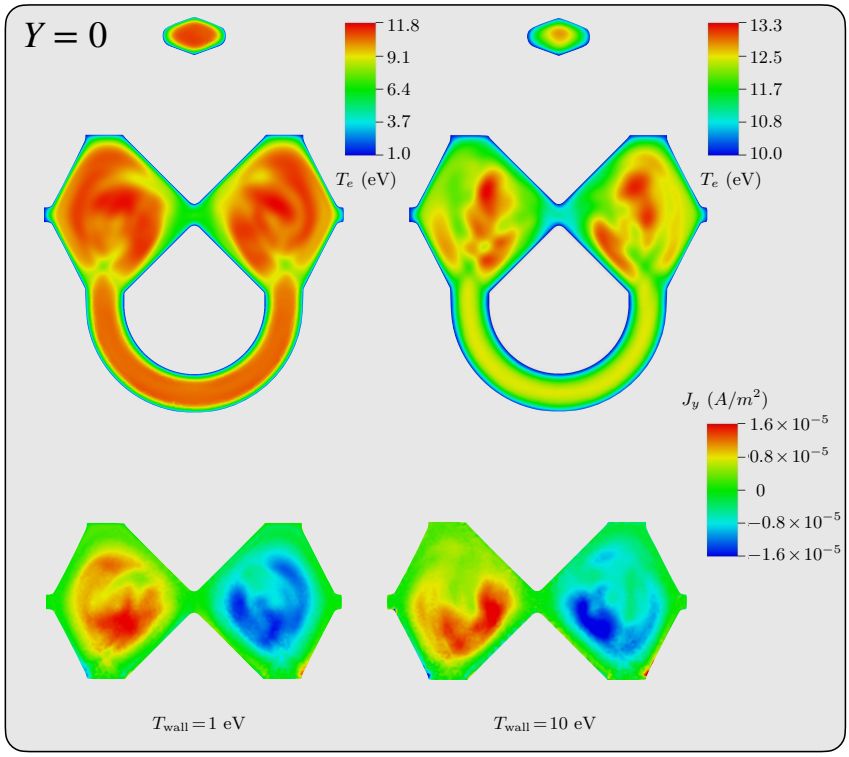


FIG. 9: Time-averaged T_e and J_y on a toroidal slice (X-Z plane) for $T_{wall}=1$ eV, 10 eV in two-temperature PSI-Tet simulations at $f_{inj}=14.5$ kHz.

perimental density fluctuation size will require a density profile in the injectors that is a more faithful representation of the experimental fueling. Further progress will require careful reproduction of the experimental waveforms and a direct model of the experimental circuit.

V. CONCLUSIONS

A two-temperature model was implemented in PSI-Tet and provides new insights into the plasma dynamics in HIT-SI simulation and experiment. Both the PSI-Tet and NIMROD two-temperature models differ from the single-temperature models through higher volume-averaged ion temperature, lower chord-averaged density, and, for the most part, axial symmetrization of the current centroid. Ion and electron temperatures are now in qualitative agreement with ion Doppler spectroscopy and initial Thomson scattering measurements. This new model further indicates injector impedance and volume-averaged temperatures scale approximately linearly with injector frequency.

Parameter scans in the Dirichlet boundary condition for temperature and density, along with simulations exploring the artificial diffusivity, lead to new physical insights into the plasma dynamics in the HIT-SI device. At higher wall temperature, significant changes to the spatial distributions of T_e and $\boldsymbol{J}$ lead to an inward shift of the current centroid. Importantly, the plasma dynamics exhibit considerable dependence on the plasma density. Performance improves with higher volume-averaged temperatures, larger toroidal current, reduced oscillations of the current centroid position, and reduced heat flux to the wall. The low density, low frequency, high power simulation indicates a sharp rise in injector power, suggesting a sudden change in the dynamics towards a higher performance regime. These simulations motivate further experimental and numerical investigation of low density parameter regimes.

Reductions in the artificial diffusivity produce signifi-

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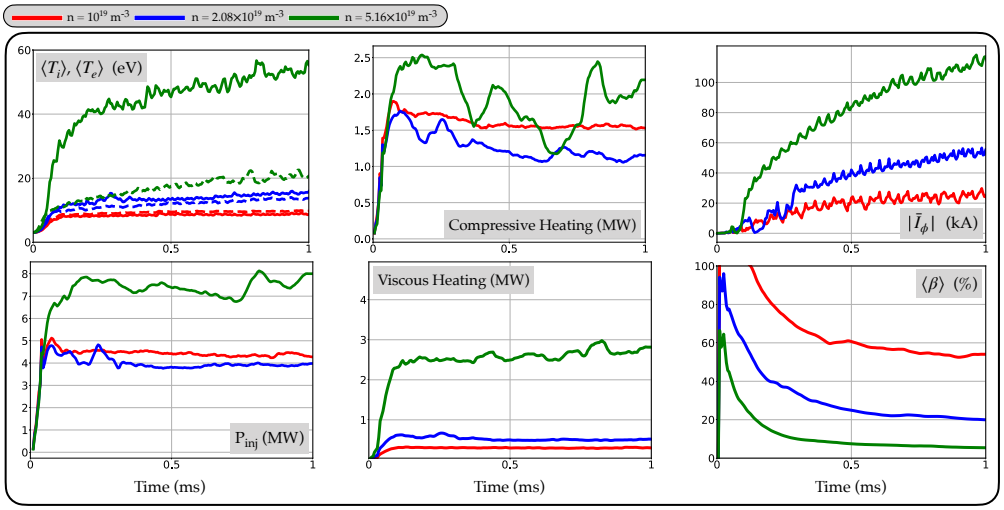


FIG. 10: Time evolution of important quantities for the wall density scan of two-temperature PSI-Tet simulations with $f_{\text{inj}} = 14.5$ kHz. $\langle T_i \rangle$ and $\langle T_e \rangle$ are in solid and dashed lines, respectively.

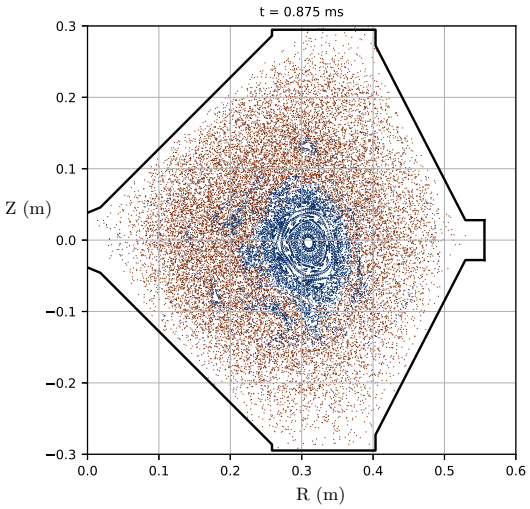


FIG. 11: Poincaré plot of $\boldsymbol{B}$ illustrates closed flux surfaces with lifetimes $50-100$ μs in a low density, high power, two-temperature PSI-Tet simulation at $f_{\text{inj}} = 14.5$ kHz. Points in blue approximately indicate the closed field lines. (Multimedia view).

cant decreases in the chord-averaged density in the device, leading to higher temperatures and toroidal currents. The absolute and relative density fluctuations increase as D decreases, improving the qualitative agreement with experiment. Further reduction may potentially lead to numerical instability near these sharp density gradients. Reduction in the artificial diffusivity in PSI-Tet may be possible by using a hyper-diffusivity term $D_h \nabla^4 n$, which can provide a larger ratio between smoothing at the grid scale and the global scale and has been successfully applied to NIMROD simulations. The observations here strongly support the claim that variations in density produce a strong impact on the dynamics.

The present work illustrated the dynamical changes in the HIT-SI device with a two-temperature model and as a number of simulation parameters vary, avoiding quantitative experimental validation because of the lack of a realistic circuit model for the injector dynamics. Future work should attempt detailed validation with the experiment with the new two-temperature model, as has been done with previous models[48]. Towards this goal, a circuit model[28] of the injectors will be implemented in PSI-Tet so that injector drive can be captured more completely by the simulation. This is expected to be important for experimental validation because the reconnection heating, propagated through the viscous heating, depends strongly on the phase between the injector waveforms when the injector reverses direction. The primary reversal phase was found in previous work[19] to last only a few μs, consistent with the Sweet-Parker[49,50] reconnection timescale. Further physical understanding and

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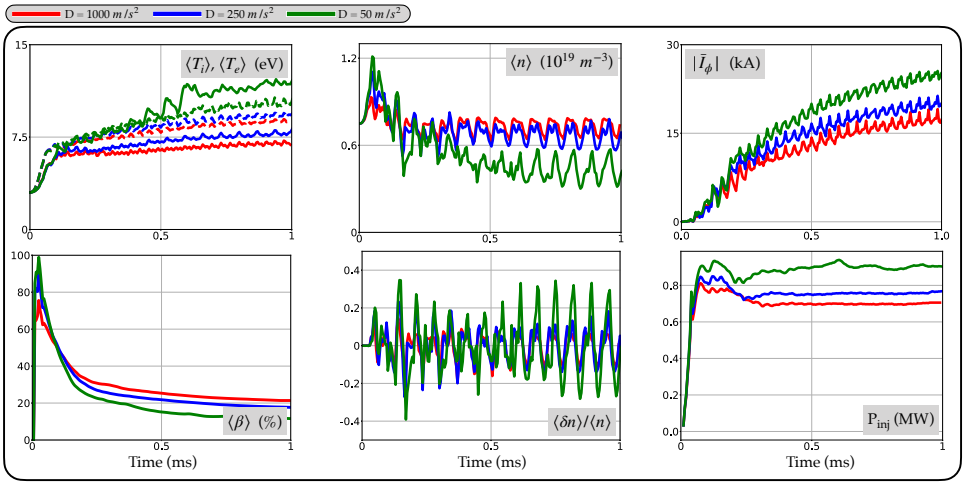


FIG. 12: Time evolution of important quantities for the artificial diffusivity scan of two-temperature PSI-Tet simulations with $f_{\text{inj}} = 14.5$ kHz. $\langle T_i \rangle$ and $\langle T_e \rangle$ are in solid and dashed lines, respectively.

improved validation with the experiment could also be obtained from an implementation of a more realistic closure for heat transport and anisotropic viscosity.

VI. ACKNOWLEDGEMENTS

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REFERENCES

[1] CR Sovinec, AH Glasser, TA Gianakon, DC Barnes, RA Nebel, SE Kruger, DD Schnack, SJ Plimpton, A Tarditi, MS Chu, et al. Nonlinear magnetohydrodynamics simulation using high-order finite elements. *Journal of Computational Physics*, 195(1):355–386, 2004.

[2] DD Schnack, DC Barnes, Z Mikic, Douglas S Harned, and EJ Caramana. Semi-implicit magnetohydrodynamic calculations. *Journal of Computational Physics*, 70(2):330–354, 1987.

[3] Wonchull Park, EV Belova, GY Fu, XZ Tang, HR Strauss, and LE Sugiyama. Plasma simulation studies using multilevel physics models. *Physics of Plasmas*, 6(5):1796–1803, 1999.

[4] VA Izzo and TR Jarboe. A numerical assessment of the Lundquist number requirement for relaxation current drive. *Physics of Plasmas*, 10(7):2903–2911, 2003.

[5] Cihan Akcay. *Extended magnetohydrodynamic simulations of the helicity injected torus (HIT-SI) spheromak experiment with the NIMROD code*. PhD thesis, University of Washington, Seattle, 2013.

[6] TR Jarboe, BS Victor, BA Nelson, CJ Hansen, C Akcay, DA Ennis, NK Hicks, AC Hossack, GJ Marklin, and RJ Smith. Imposed-dynamo current drive. *Nuclear Fusion*, 52(8):083017, 2012.

[7] KD Morgan, TR Jarboe, AC Hossack, RN Chandra, and CJ Everson. Validation of extended magnetohydrodynamic simulations of the HIT-SI3 experiment using the NIMROD code. *Physics of Plasmas*, 24(12):122510, 2017.

[8] Jeffrey P Freidberg. *Ideal MHD*. Cambridge University Press, 2014.

[9] Thomas R Jarboe. Review of spheromak research. *Plasma Physics and Controlled Fusion*, 36(6):945, 1994.

[10] Paul Murray Bellan. Spheromaks. a practical application of magnetohydrodynamic dynamos and plasma self-organization. 2000.

[11] T. E. Benedett, C. J. Hansen, K. D. Morgan, and T. R. Jarboe. Effects of temperature and density evolution in MHD simulations of HIT-SI. *Physics of Plasmas*, 27(4):042508, 2020.

[12] Christopher James Hansen. *MHD modeling in complex 3D geometries: Towards predictive simulation of SIHI current drive*. PhD thesis, 2014.

[13] Aaron Hossack. *A study of plasma dynamics in HIT-SI using ion Doppler spectroscopy*. PhD thesis, University of Washington, Seattle, 2015.

[14] Chris Everson. HIT-SI3 Thomson scattering results. *Bulletin of the American Physical Society*, 2019.

[15] TR Jarboe, WT Hamp, GJ Marklin, BA Nelson, RG O'Neill, AJ Redd, PE Sieck, RJ Smith, and JS Wrobel. Spheromak formation by steady inductive helicity injection. *Physical review letters*, 97(11):115003, 2006.

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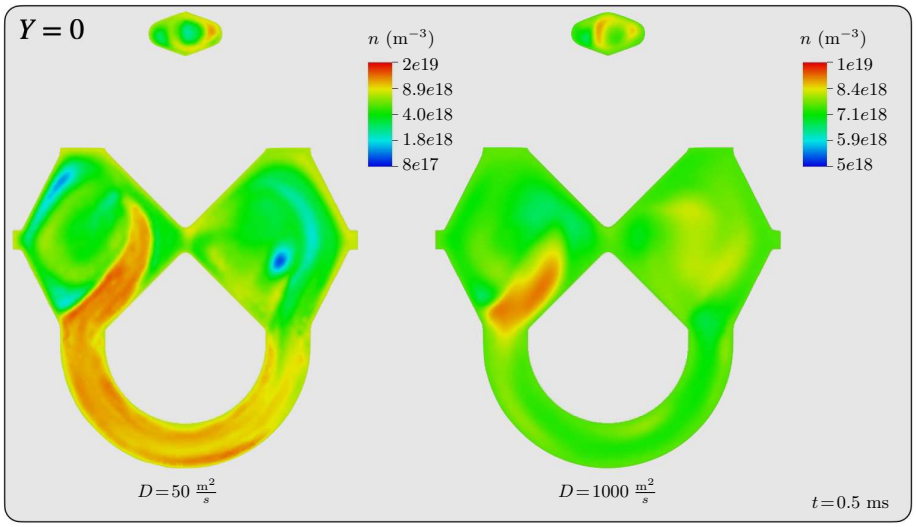


FIG. 13: Comparison of the 2D spatial density profiles for $D=50$ m$^2/s$ and $D=1000$ m$^2/s$ in two-temperature PSI-Tet simulations at $f_{\text{inj}}=14.5$ kHz. The contours indicate that pockets of low density form as D decreases. Contours also show significantly higher density due to compression in the injector regions.

[16]TR Jarboe, CJ Hansen, AC Hossack, GJ Marklin, KD Morgan, BA Nelson, DA Sutherland, and BS Victor. A proof of principle of imposed dynamo current drive: Demonstration of sufficient confinement. *Fusion Science and Technology*, 66(3):369–384, 2014.

[17]Jonathan Scott Wrobel. *A study of HIT-SI plasma dynamics using surface magnetic field measurements*. University of Washington, 2011.

[18]Kyle Morgan. *Finite-beta simulations of HIT-SI and HIT-SI3 using the NIMROD code*. PhD thesis, University of Washington, Seattle, 2018.

[19]C Hansen, G Marklin, B Victor, C Akcay, and T Jarboe. Simulation of injector dynamics during steady inductive helicity injection current drive in the HIT-SI experiment. *Physics of Plasmas*, 22(4):042505, 2015.

[20]E B Hooper, T A Kopriva, B I Cohen, D N Hill, H S McLean, R D Wood, S Woodruff, and C R Sovinec. Magnetic reconnection during flux conversion in a driven spheromak. *Physics of Plasmas*, 12(9):092503, 2005.

[21]J B O'Bryan and C R Sovinec. Simulated flux-rope evolution during non-inductive startup in Pegasus. *Plasma Physics and Controlled Fusion*, 56(6):064005, 2014.

[22]SI Braginskii and MA Leontovich. Reviews of plasma physics, 1965.

[23]Lyman Spitzer. *Physics of fully ionized gases*. Courier Corporation, 2006.

[24]Joseph D Huba. NRL plasma formulary. Technical report, Naval Research Lab Washington DC plasma physics division, 2006.

[25]T D Rognlien and M E Rensink. Edge-plasma models and characteristics for magnetic fusion energy devices. *Fusion Engineering and Design*, 60(4):497–514, 2002.

[26]E T Meier, R J Goldston, E G Kaveeva, M A Makowski, S Mordijck, V A Rozhansky, I Yu Senichenkov, and S P Voskoboynikov. Analysis of drift effects on the tokamak power scrape-off width using SOLPS-ITER. *Plasma Physics and Controlled Fusion*, 58(12):125012, 2016.

[27]D.A. Sutherland. *Measurements of neutral particles and simulations of plasma-neutral dynamics in the HIT-SI3 experiment*. PhD thesis, University of Washington, Seattle, 2019.

[28]E Bickford Hooper, Bruce I Cohen, HS McLean, RD Wood, CA Romero-Talamas, and CR Sovinec. NIMROD resistive magnetohydrodynamic simulations of spheromak physics. *Physics of Plasmas*, 15(3):032502, 2008.

[29]Kyle Morgan, Thomas Jarboe, and Cihan Akcay. Formation of closed flux surfaces in spheromaks sustained by steady inductive helicity injection. *Nuclear Fusion*, 59(6):066037, 2019.

[30]Jacob R King, Keith H Burrell, Andrea M Garofalo, Richard J Groebner, SE Kruger, Alexei Y Pankin, and Philip B Snyder. NIMROD modeling of quiescent H-mode: reconstruction considerations and saturation mechanism. *Nuclear Fusion*, 57(2):022002, 2016.

[31]Charlson C Kim, Yueqiang Liu, Paul B Parks, Lang L Lao, Michael Lehnen, and Alberto Loarte. Shattered pellet injection simulations with NIMROD. *Physics of Plasmas*, 26(4):042510, 2019.

[32]Alexander N. Brooks and Thomas J.R. Hughes. Streamline upwind/Petrov-Galerkin formulations for convection dominated flows with particular emphasis on the incompressible Navier-Stokes equations. *Computer Methods in Applied Mechanics and Engineering*, 32(1):199 – 259, 1982.

[33]Hassan K Khalil. Nonlinear systems. *Upper Saddle River*, 2002.

[34]J. S. Wrobel, C. J. Hansen, T. R. Jarboe, R. J. Smith, A. C. Hossack, B. A. Nelson, G. J. Marklin, D. A. Ennis, C. Akcay, and B. S. Victor. Relaxation-time measurement via a time-dependent helicity balance model. *Physics of Plasmas*, 20(1):012503, 2013.

[35]Y. Ono, H. Tanabe, Y. Hayashi, T. Ii, Y. Narushima, T. Yamada, M. Inomoto, and C. Z. Cheng. Ion and electron heating characteristics of magnetic reconnection in a two flux loop merging experiment. *Phys. Rev. Lett.*, 107:185001, Oct 2011.

[36]G Fiksel, AF Almagri, BE Chapman, VV Mirnov, Y Ren, JS Sarff, and PW Terry. Mass-dependent ion heating during magnetic reconnection in a laboratory plasma. *Physical review*

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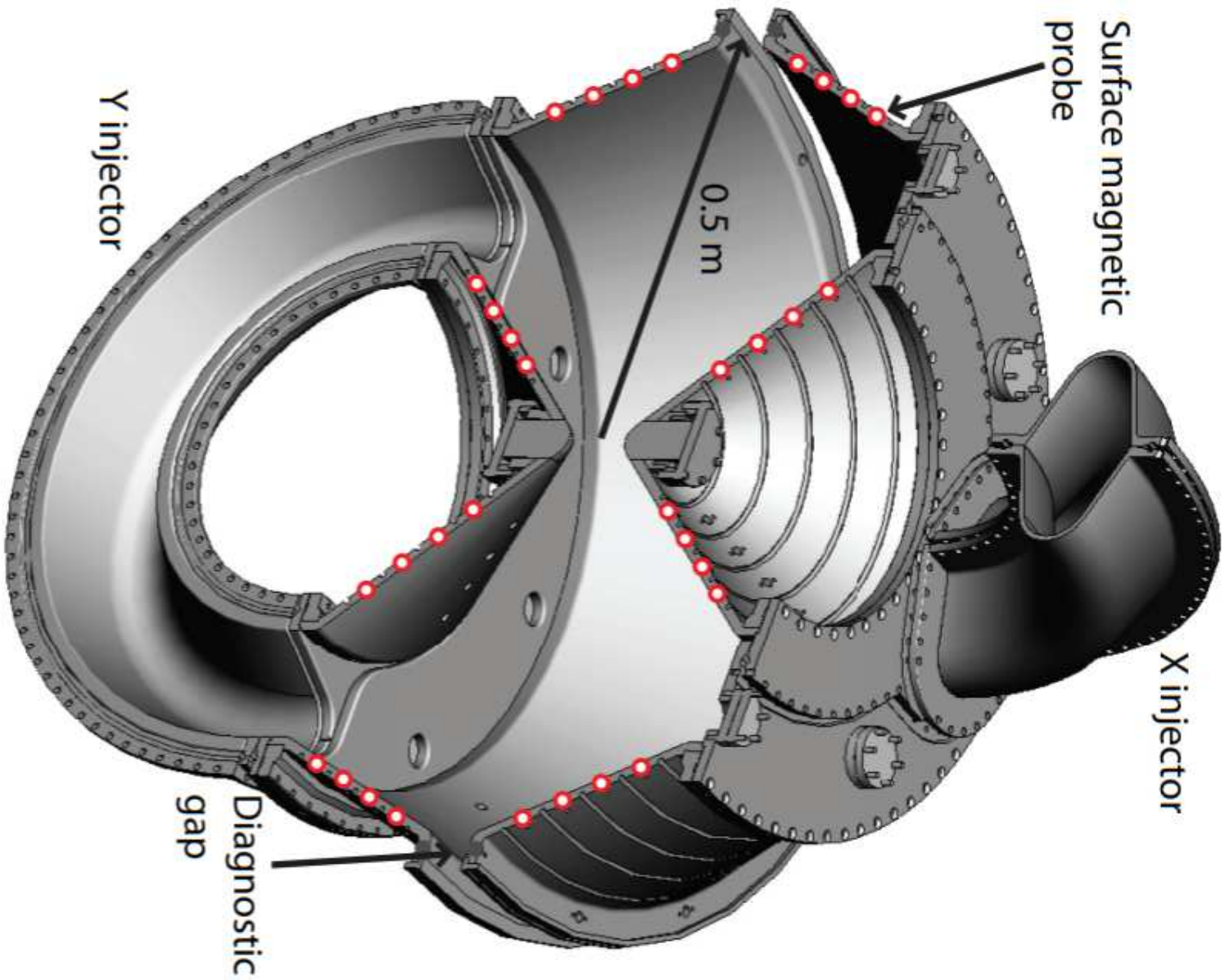
letters, 103(14):145002, 2009.
[37]BS Victor, TR Jarboe, CJ Hansen, C Akcay, KD Morgan, AC Hossack, and BA Nelson. Sustained spheromaks with ideal n=1 kink stability and pressure confinement. *Physics of Plasmas*, 21(8):082504, 2014.
[38]TR Jarboe, C Akcay, MA Chilenski, DA Ennis, CJ Hansen, NK Hicks, RZ Aboul Hosn, AC Hossack, GJ Marklin, BA Nelson, et al. Recent results from the HIT-SI experiment. *Nuclear Fusion*, 51(6):063029, 2011.
[39]RG O'Neill, GJ Marklin, TR Jarboe, C Akcay, WT Hamp, BA Nelson, AJ Redd, RJ Smith, BT Stewart, JS Wrobel, et al. A fully relaxed helicity balance model for an inductively driven spheromak. *Physics of Plasmas*, 14(11):112304, 2007.
[40]Peter J Schmid. Dynamic mode decomposition of numerical and experimental data. *Journal of fluid mechanics*, 656:5–28, 2010.
[41]J. H. Tu, C. W. Rowley, D. M. Luchtenburg, S. L. Brunton, and J. N. Kutz. On dynamic mode decomposition: theory and applications. *Journal of Computational Dynamics*, 1(2):391–421, 2014.
[42]Steven L Brunton and J Nathan Kutz. *Data-driven science and engineering: Machine learning, dynamical systems, and control*. Cambridge University Press, 2019.
[43]A. A. Kaptanoglu, K. D. Morgan, C. J. Hansen, and S. L. Brunton. Characterizing magnetized plasmas with dynamic mode decomposition. *Physics of Plasmas*, 27(3):032108, 2020.
[44]Aaron C Hossack, Taylor Firman, Thomas R Jarboe, James R Prager, Brian S Victor, Jonathan S Wrobel, and Timothy Ziemba. Reduction of plasma density in the helicity injected torus with steady inductance experiment by using a helicon pre-ionization source. *Review of Scientific Instruments*, 84(10):103506, 2013.
[45]D.A. Ennis, B.S. Victor, J.S. Wrobel, C. Akcay, T.R. Jarboe, G.J. Marklin, B.A. Nelson, and R.J. Smith. New understandings and achievements from independent-injector drive experiments on HIT-SI. *Nuclear Fusion*, 50(7):072001, jun 2010.
[46]Claude Mercier. On a representation of toroidal surfaces. applications to magnetohydrodynamic equilibria. *Nuclear Fusion*, 3(2):89, 1963.
[47]RM Mayo and GJ Marklin. Numerical calculation of Mercier beta limits in spheromaks. *The Physics of fluids*, 31(6):1812–1815, 1988.
[48]C Hansen, B Victor, K Morgan, T Jarboe, A Hossack, G Marklin, BA Nelson, and D Sutherland. Numerical studies and metric development for validation of magnetohydrodynamic models on the HIT-SI experiment. *Physics of Plasmas*, 22(5):056105, 2015.
[49]PA Sweet. Electromagnetic phenomena in cosmical physics (IAU Symp. 6), ed, 1958.
[50]Eugene N Parker. Sweet's mechanism for merging magnetic fields in conducting fluids. *Journal of Geophysical Research*, 62(4):509–520, 1957.

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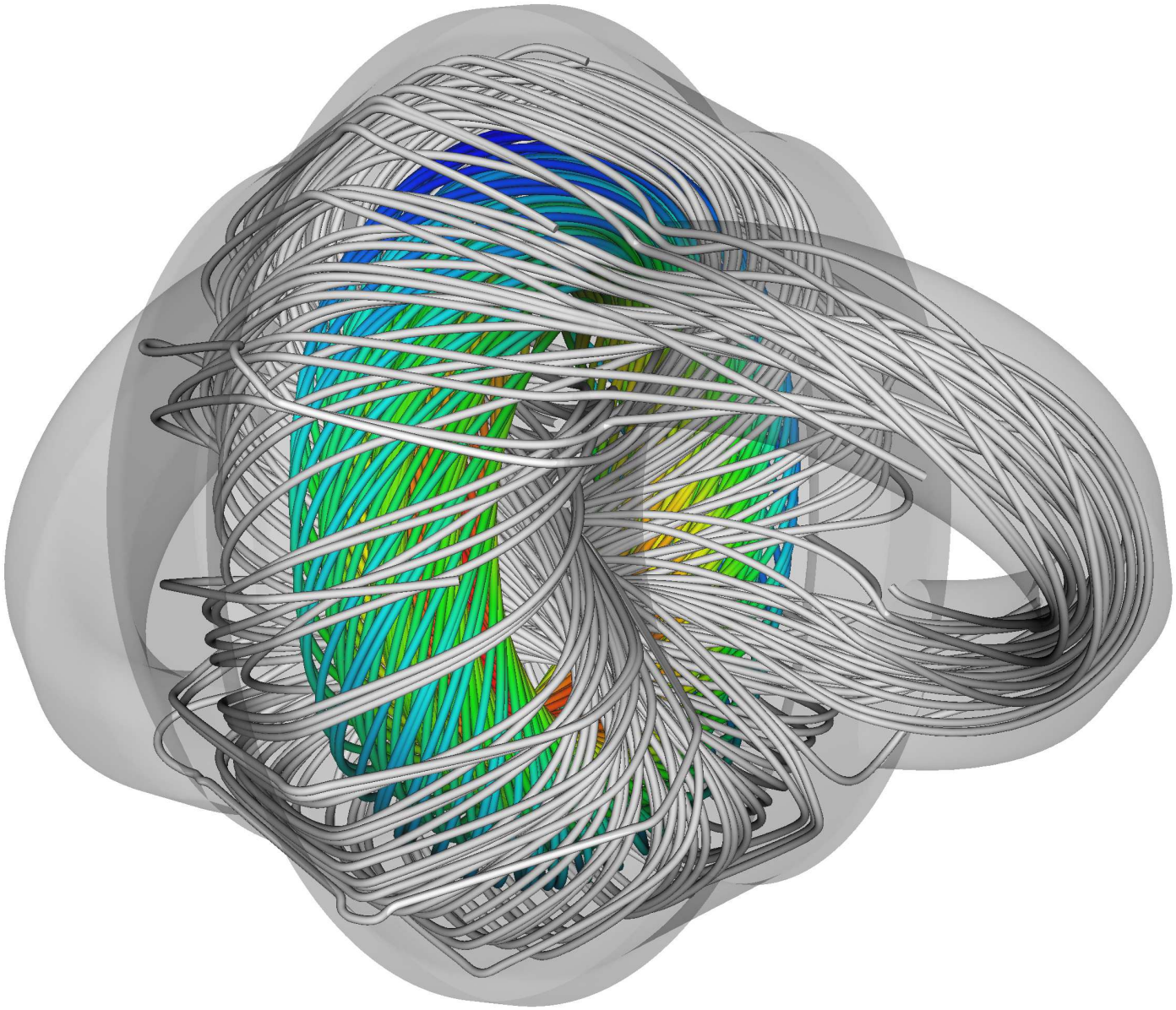
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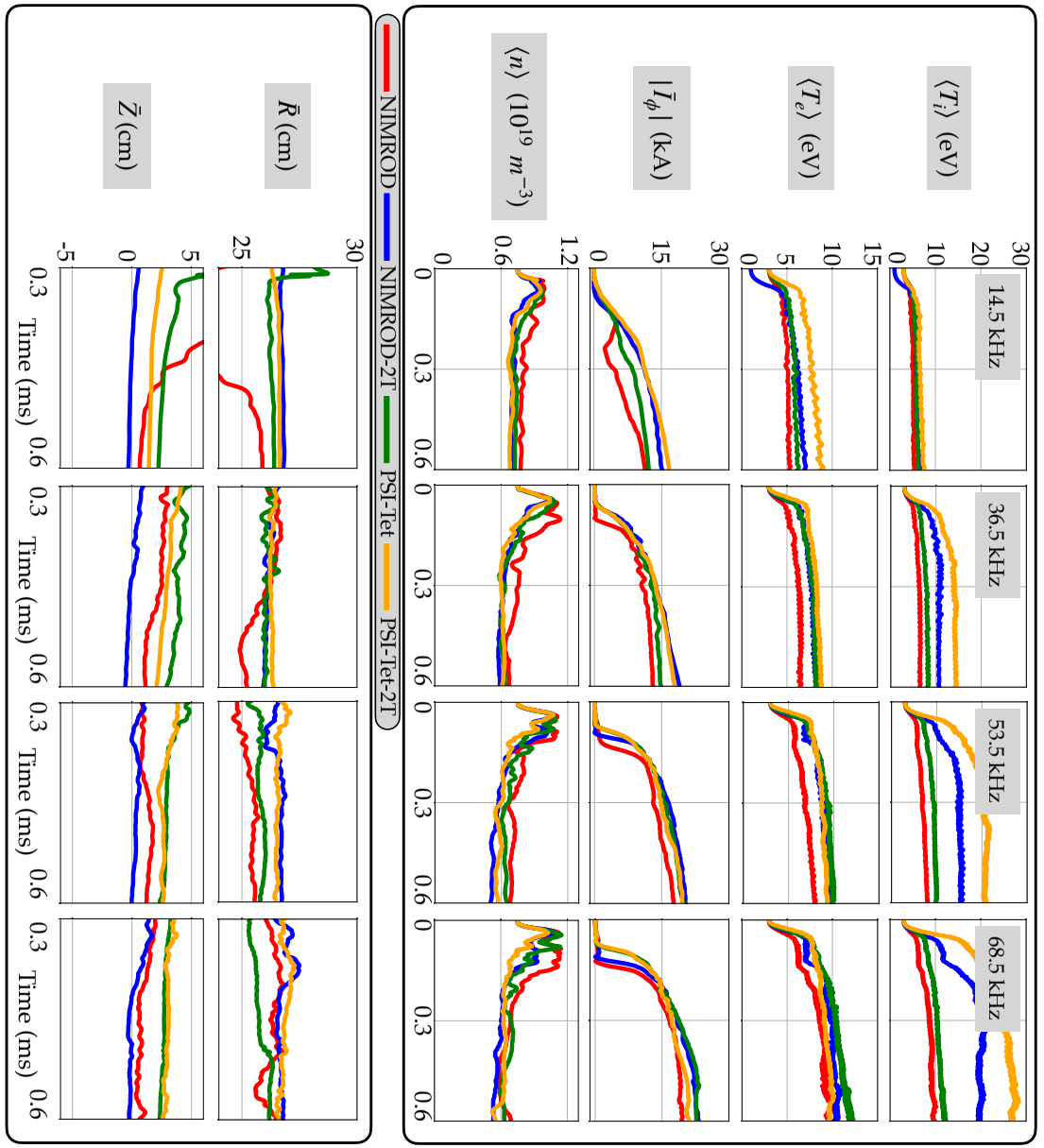
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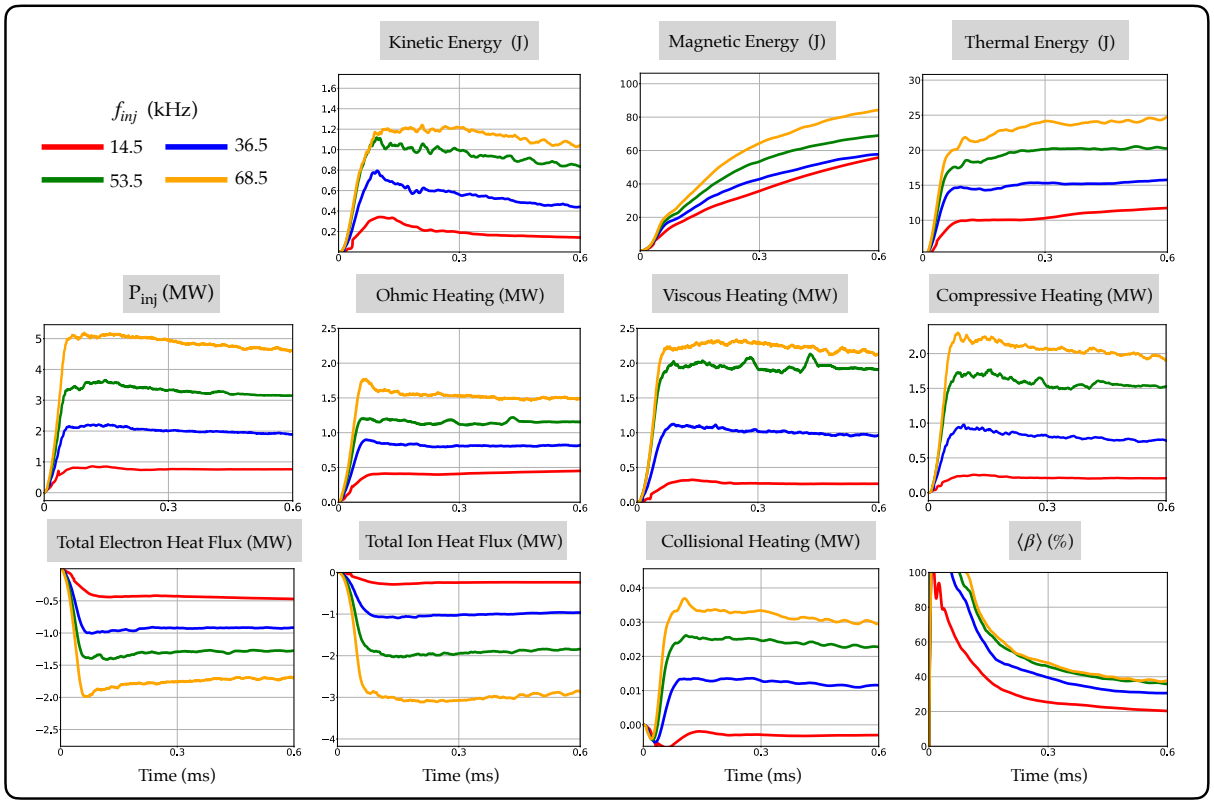
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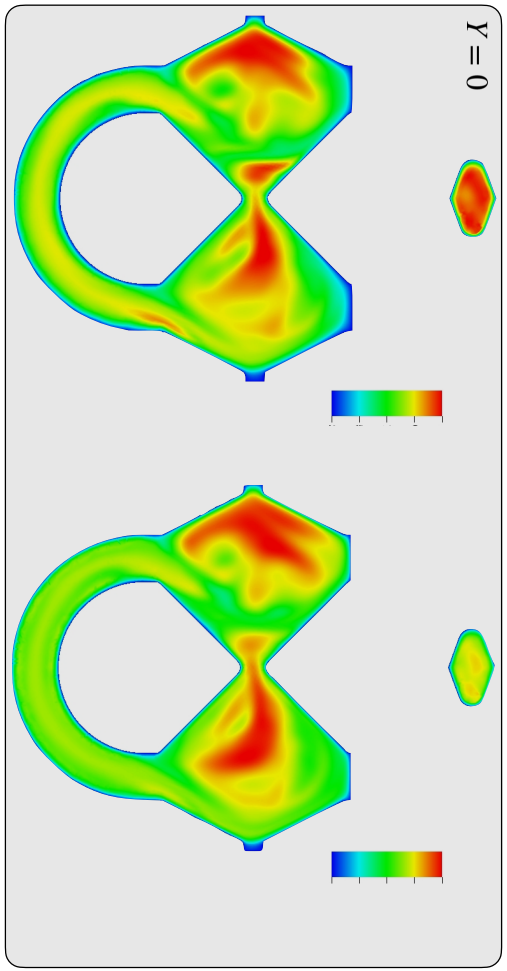
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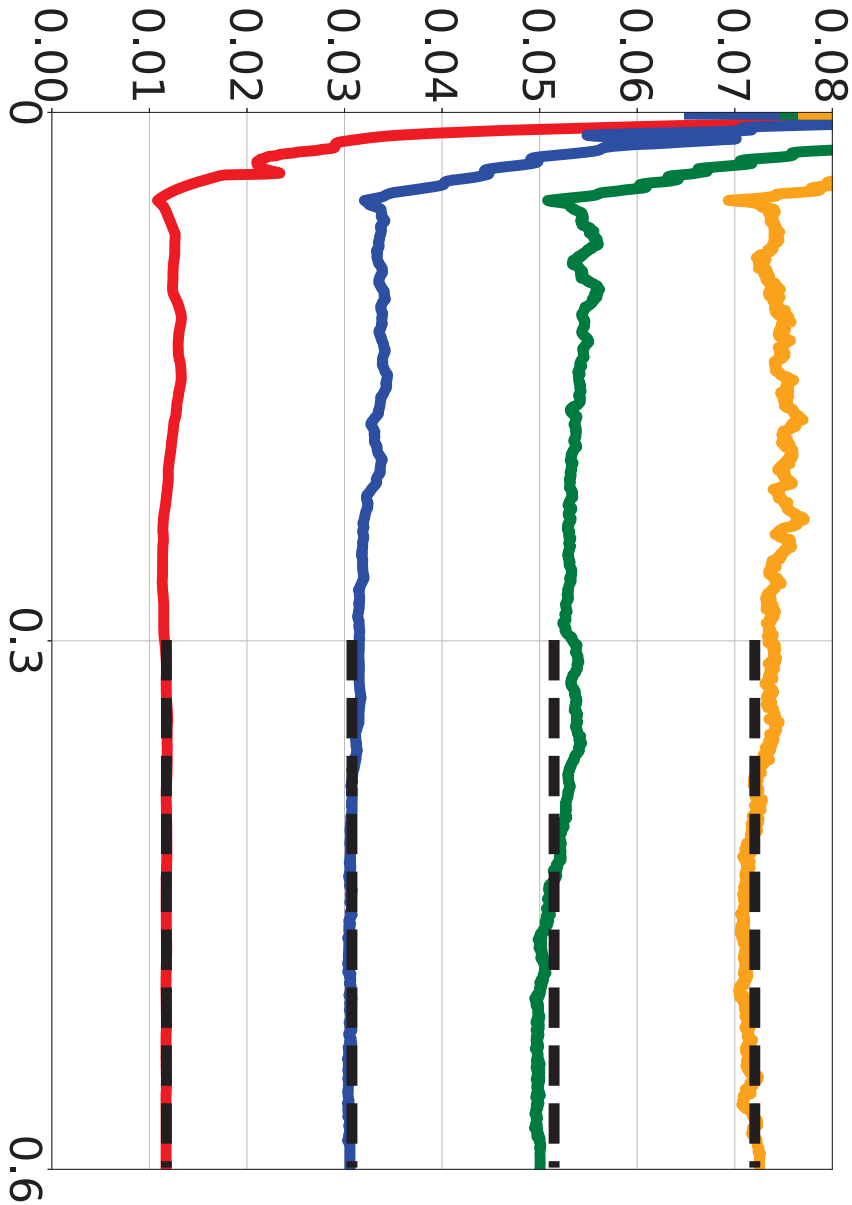
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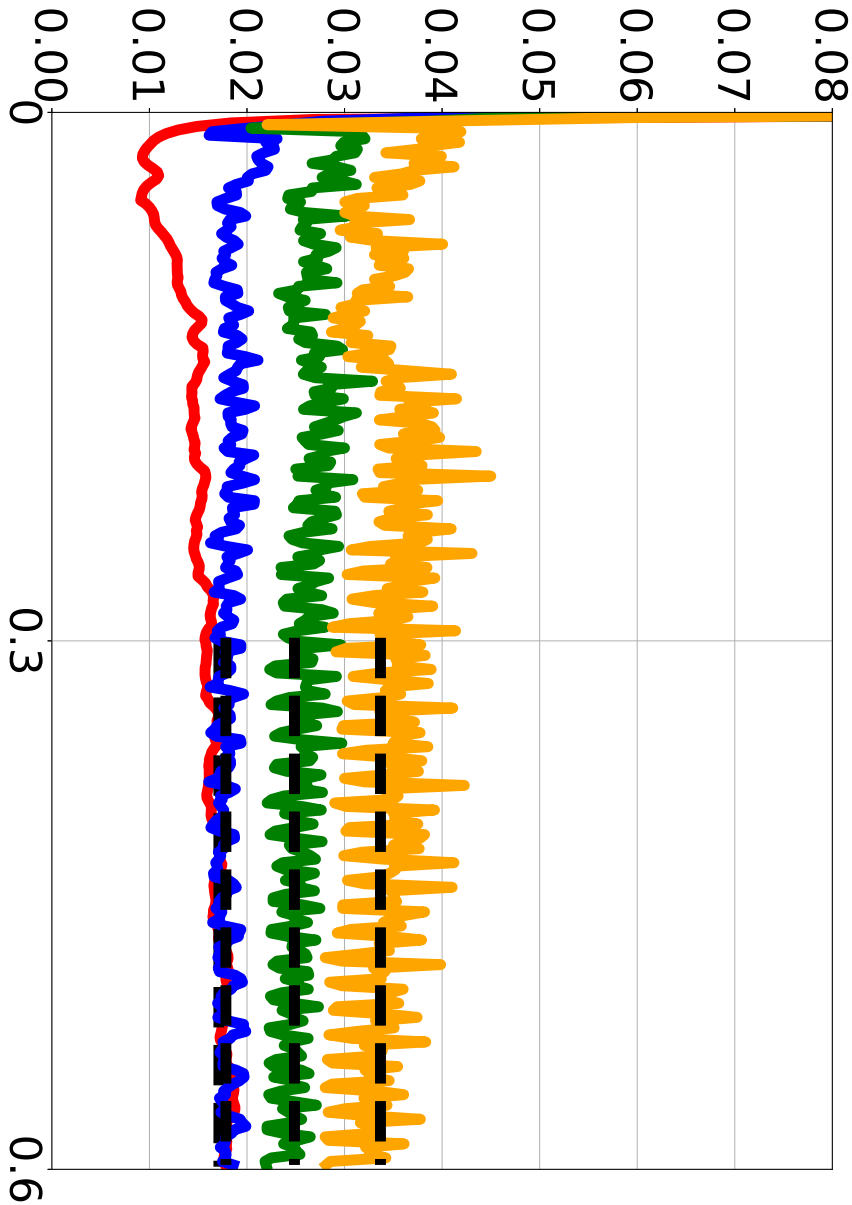
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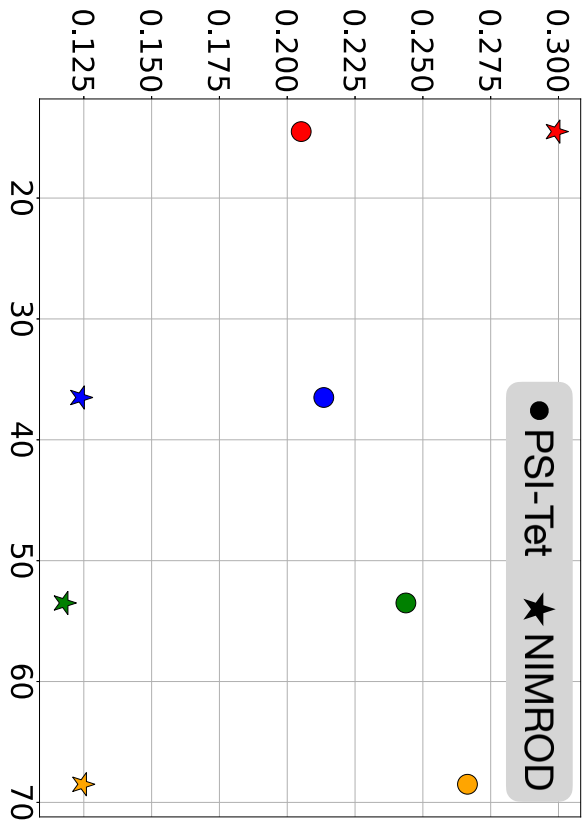
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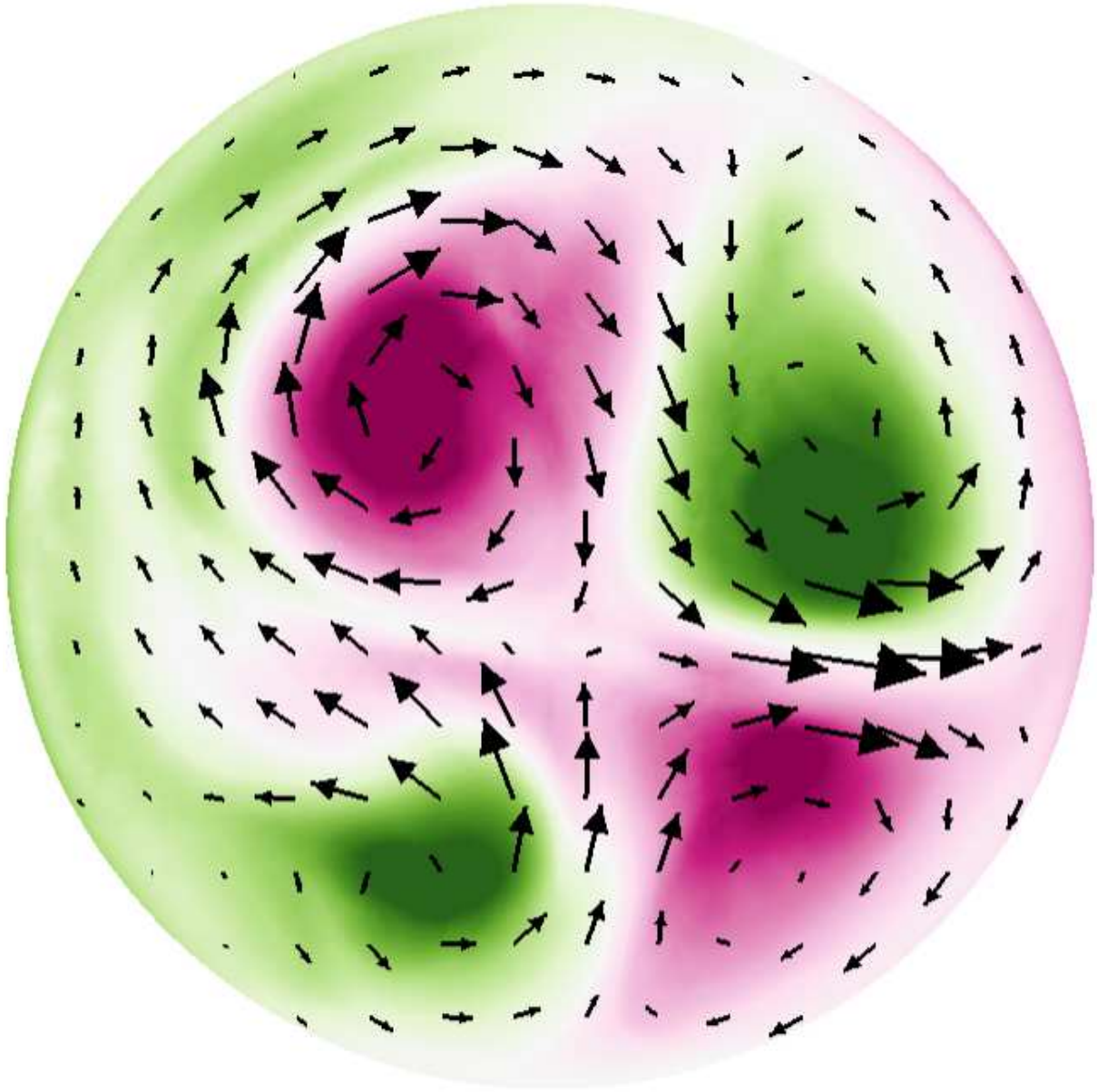
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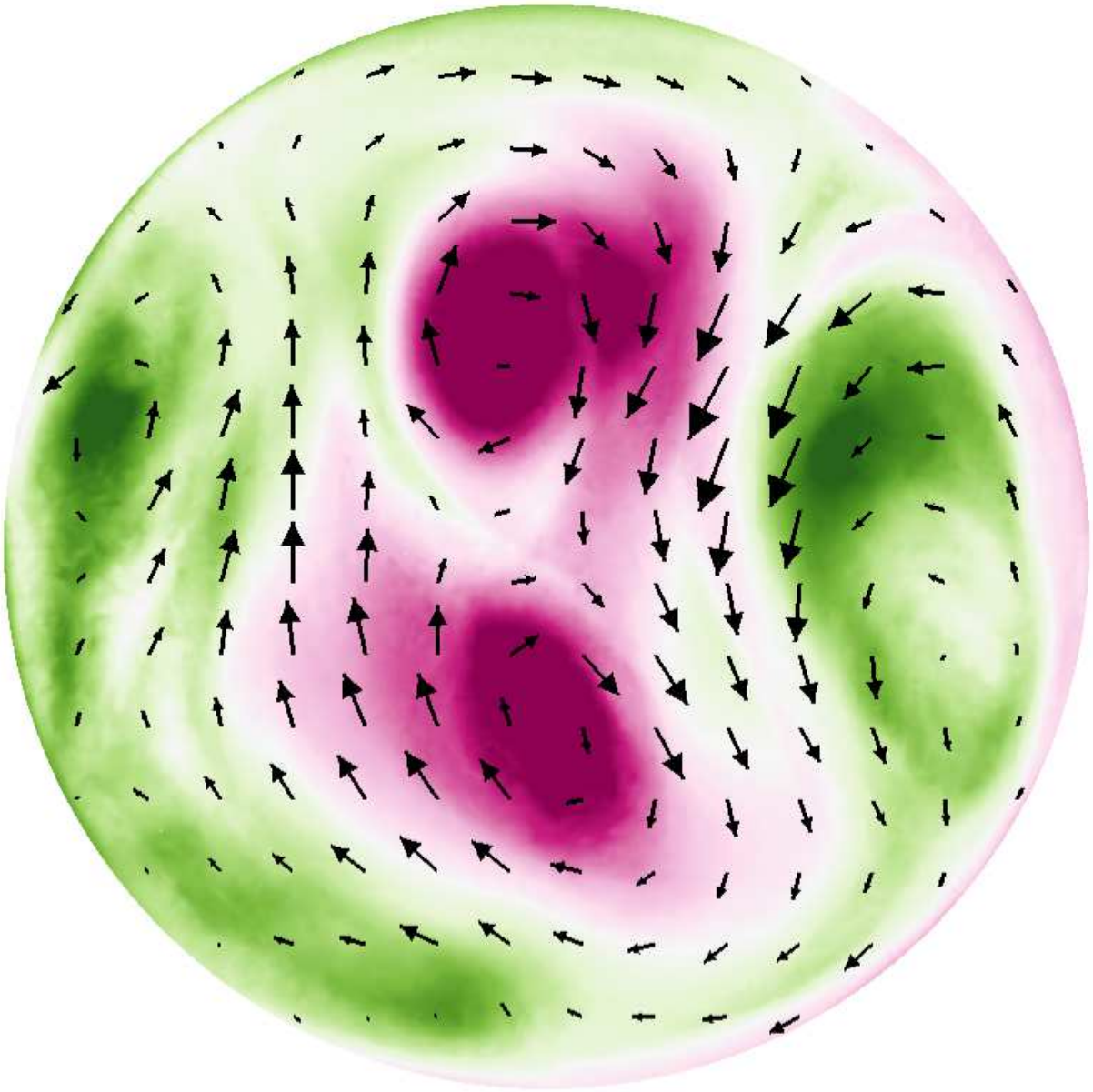
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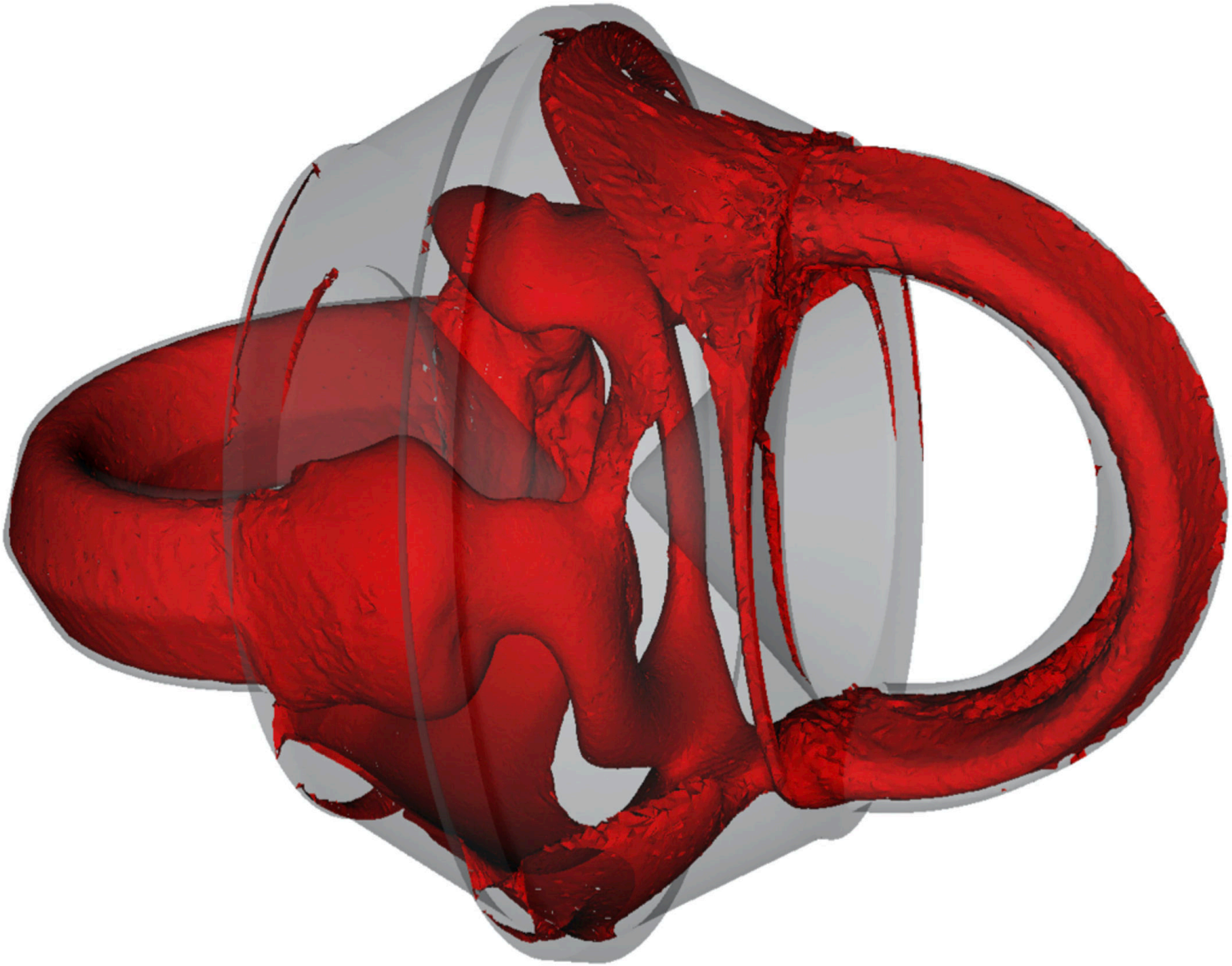
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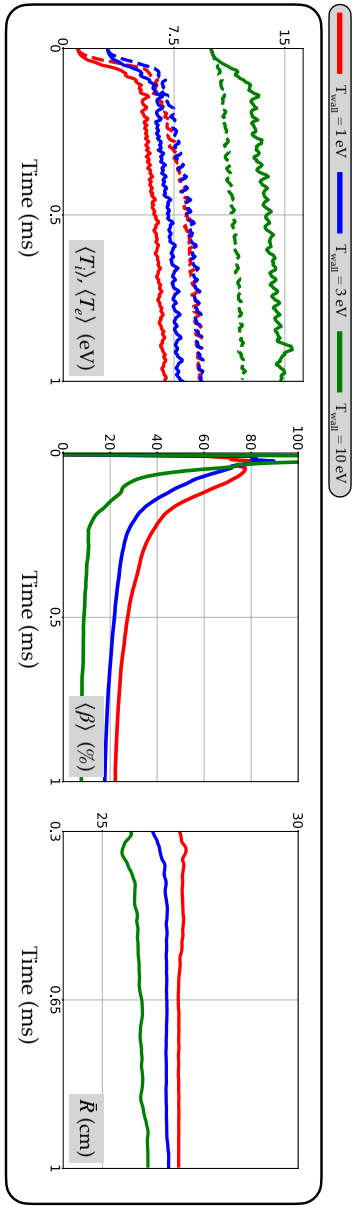
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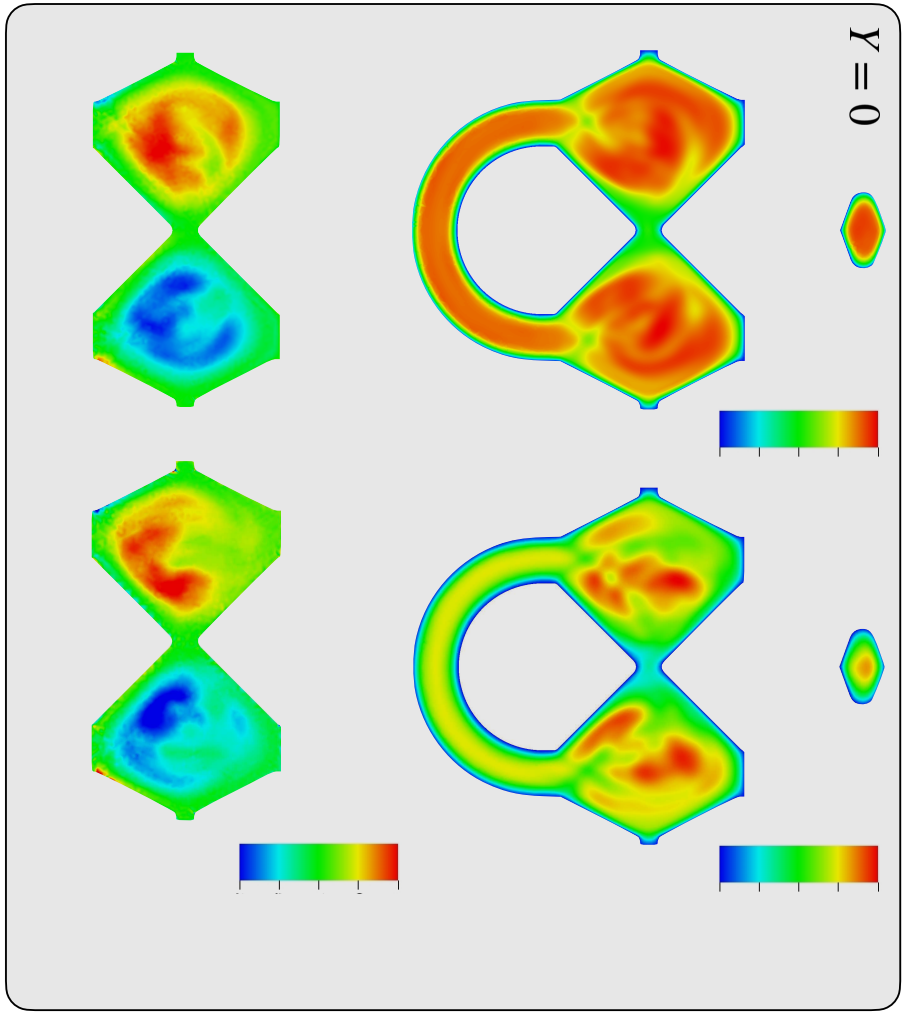
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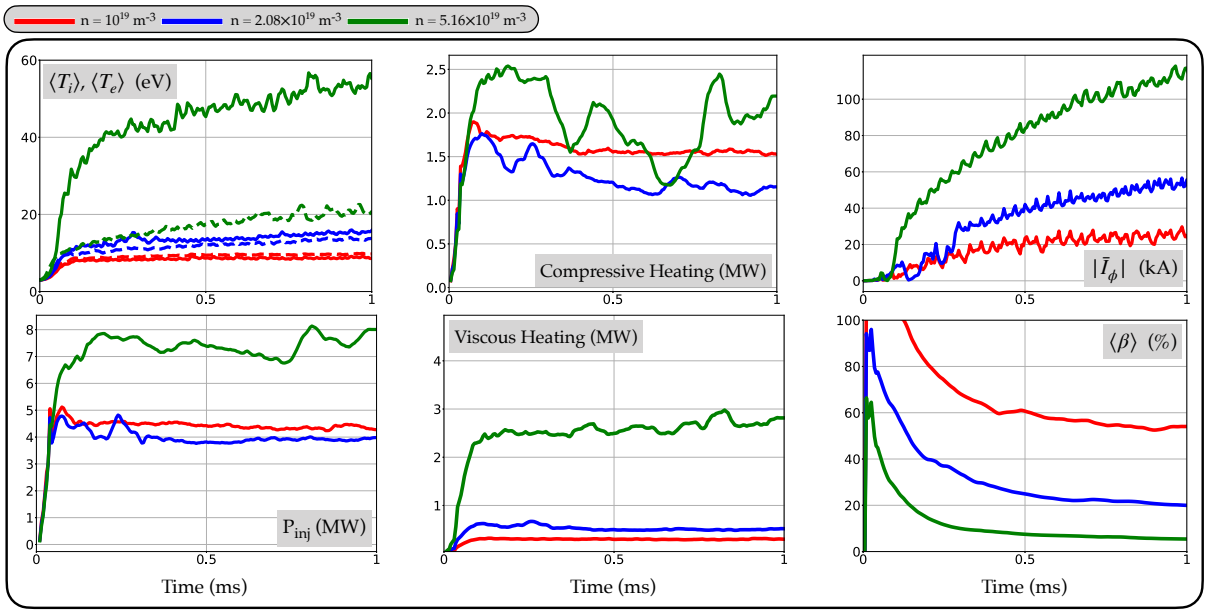
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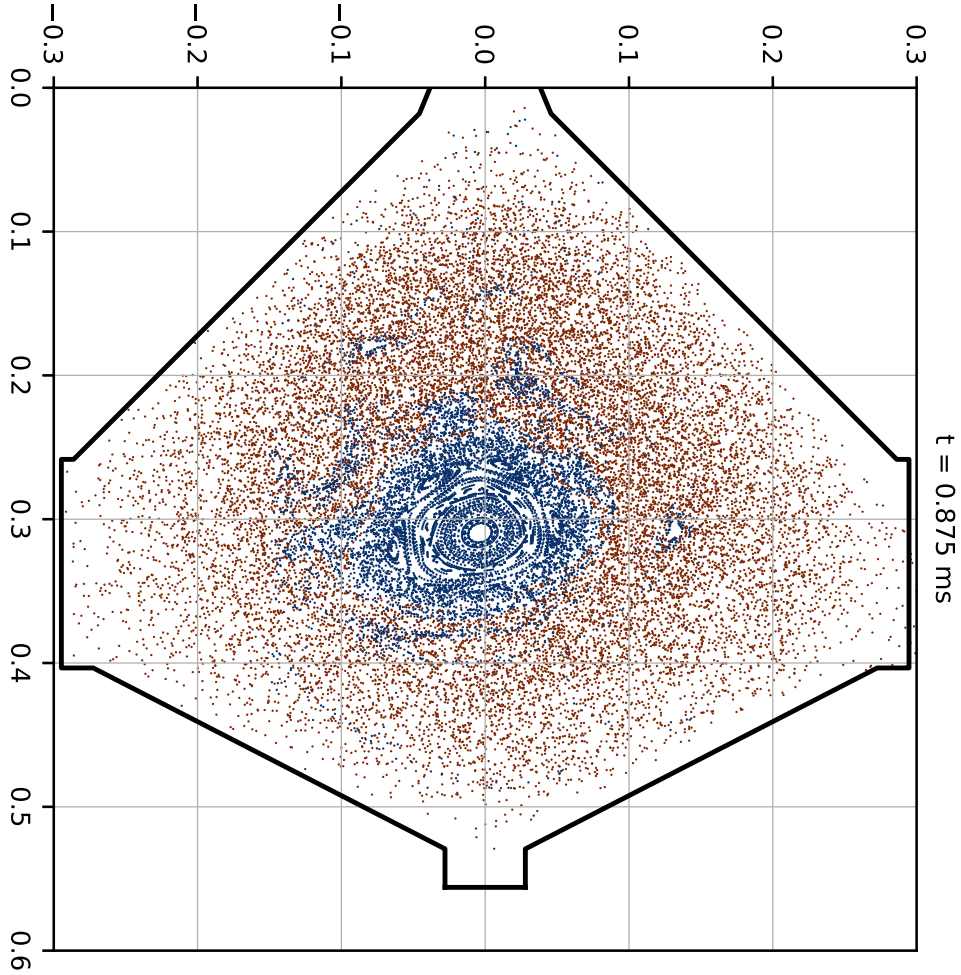
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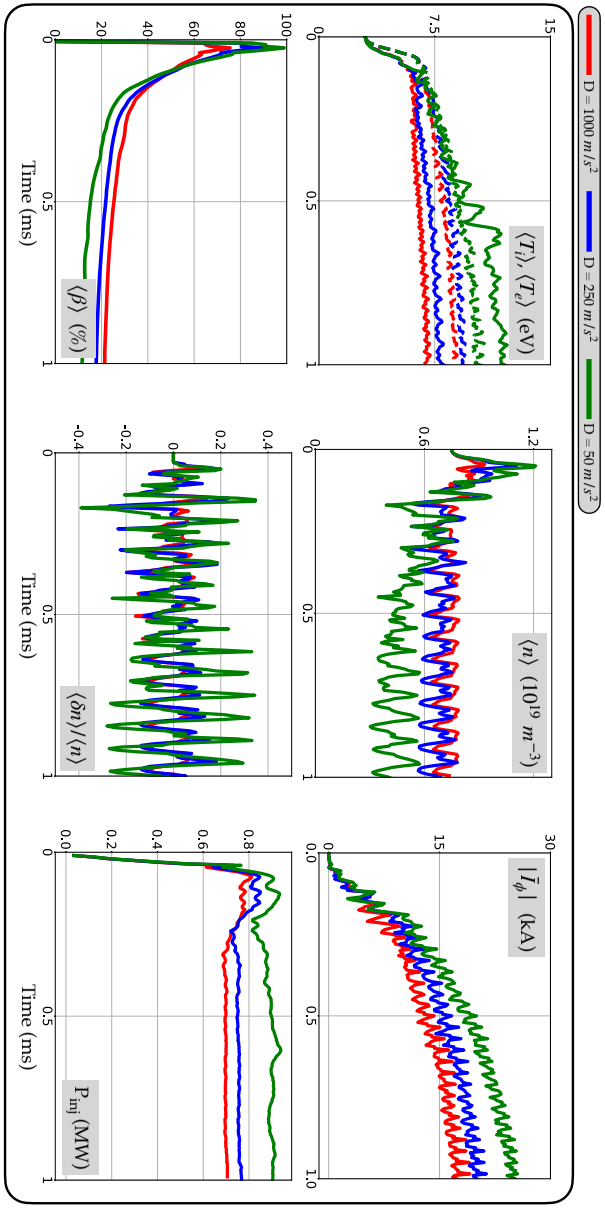
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